

Adaptive Fully Deniable Encryption from Weaker Lattice Assumptions

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Abstract. Deniable Encryption (Canetti et al., CRYPTO 1997) has been a promising direction to achieve an ideal private communication, which provides security even under coercion of the parties, and this attempt has been continuous as from building Bi-deniability (O’Neill et al., CRYPTO 2011) to post-quantum protocols (Agrawal et al., CRYPTO 2021), which are secure only if one of their parties gets coerced. The peak of advancements was the proposition of a Fully Deniable Encryption protocol (Canetti et al., CRYPTO 2020), where even if both the sender and receiver get coerced, the content still remains secure. However, Canetti et al. limit the Fully Deniable Encryption to post-execution coercion and identify adaptive coercion, where parties can be coerced during protocol execution, as unaddressed. Also, their construction, not only is vulnerable to quantum attacks, but also requires subexponentially secure primitives, and building Fully Deniable Encryption from weaker assumptions remains an open problem. Additionally, the lack of explicit implications in the aforementioned works makes the process of extracting their Deniable Encryption applications slower.

In this work, we fill these gaps by constructing a Fully Deniable Encryption that provides adaptive coercion in post-quantum settings with polynomially secure primitives. Moreover, we introduce the notion of Strong Off-the-record Deniability, where both parties lie, but their claims are consistent; we call a protocol that satisfies Bideability and Strong Off-the-record Deniability as Strongly Fully Deniable Encryption. Also, unlike most prior works, we actually show the applicability of this protocol by putting it into a practical scheme and proving that the resulting system achieves coercion-resistance.

Keywords: Applied Cryptography, Privacy-preserving Protocols, Lattice-based Cryptography, Electronic Voting, Coercion Resistance,

1 Introduction

The ability to communicate secretly over public channels without any previously shared secrets is, indeed, one of the fundamental principles of contemporary cryptography [26]. Nonetheless, traditional concepts of secure communication provide secrecy only if the internal states of the parties remain concealed [21], and it has been proven that non-interactive non-committing encryption is impossible [31]. In case either the sender or the receiver is later forced or given money to disclose their secret keys and randomness, conventional encryption methods will not protect them [4]. Additionally, in multiple protocols, the communication log practically binds the parties to the

disclosed message, thus creating a situation where asserting that another message was sent becomes non-credible.

Deniable encryption, introduced by Canetti, Dwork, Naor, and Ostrovsky [9], addresses this limitation by enabling parties to produce fake internal states that make any transcript appear consistent with any plaintext of their choice. Since its emergence, deniable encryption has shown promising real-world applications from Signal Messenger [15] to VeraCrypt [1, 3] and MobiWear for smartwatches [13]. The defining feature of deniable encryption is that an adversary cannot distinguish whether the parties revealed their true randomness and plaintext or provided fabricated alternatives [9, 11]. This guarantee holds even when the adversary observes the entire communication and later coerces the parties.

The development of fully deniable encryption has proceeded incrementally through a series of partial results. The first proposal [9] needed ciphertexts of superpolynomial lengths to get a negligibly small distinguishing advantage. Afterward, mainly Sahai and Waters [37] made progress on this by introducing the first sender-deniable encryption scheme that works with polynomial-length ciphertexts and is based on indistinguishability obfuscation. Their scheme can easily be converted to receiver-deniable encryption through a generic transformation [9] that adds one more round, making it a three-message protocol. In addition, it is possible to further extend handling adversaries who are coercing either the sender or the receiver, but not both at the same time.

The question of constructing bideniable encryption secure against simultaneous coercion of both parties remained open for nearly two decades following the initial work of Canetti et al. [9], with only partial results addressing restricted scenarios [32] and impossibility results for two-message schemes [5]. The problem has been recently solved in a positive way by Canetti, Park, and Poburinnaya [10] who introduced the first-ever completely deniable interactive encryption protocol. Among other things, their protocol realizes both bideniability and a novel security concept referred to as off-the-record deniability, which ensures the protection of the parties in situations when their declared internal states are at odds with each other. The creation of such an encryption scheme is done through three rounds of interaction, assumes the existence of indistinguishability obfuscation and one-way functions that are subexponentially secure, and is performed in the common reference string model.

Despite this substantial progress, two big gaps remain. First, the construction of [10] addresses only post-execution coercion, where adversaries coerce parties after the protocol completes. The more challenging setting of adaptive coercion, where parties may be coerced at any point during protocol execution, was explicitly identified as an open problem in [10]. Second, the reliance on subexponentially secure primitives raises the question of whether fully deniable encryption can be constructed from weaker, polynomially secure assumptions. Additionally, while deniable encryption has clear applications to coercion-resistant systems, explicit constructions and security analyses for practical schemes have been limited.

1.1 Contributions

This work resolves these open questions by constructing an adaptively secure fully deniable encryption scheme from polynomially secure assumptions, and by demonstrating its applicability to coercion-resistant electronic voting.

- We present the first adaptively secure fully deniable encryption protocol that achieves bideniability and off-the-record deniability under coercion at any point during protocol execution, relying only on polynomially secure primitives rather than subexponential assumptions.
- We introduce Strong Off-the-Record Deniability (SORD), a strengthening that requires indistinguishability between cases where both parties lie with consistent claims versus inconsistent claims, and prove our protocol satisfies this stronger notion through round-specific faking algorithms that maintain consistency with partial transcripts. SORD is resistant to the accumulating attack [5] polynomially.
- Then, we construct a coercion-resistant variant of the Belenios¹ voting system built from strongly fully deniable encryption, and prove it satisfies JCJ coercion resistance [29], providing the first explicit construction with security analysis demonstrating how fully deniable encryption yields practical coercion-resistant systems.
- Furthermore, our findings show that it is impossible to achieve strong off-the-record deniability in a voting system while preserving tally’s universal verifiability.

1.2 Related Works

As Figure 1 demonstrates, a summary of the evolution direction of protocols, the concept of deniable encryption was introduced by Canetti, Dwork, Naor, and Ostrovsky in their seminal work [9], which established the foundational definitions and provided initial constructions. However, their original scheme achieved only an inverse polynomial distinguishing advantage, requiring superpolynomially long ciphertexts to obtain security. This limitation persisted for nearly two decades, during which the community sought more efficient constructions and explored the boundaries of what could be achieved.

Sahai and Waters [37] achieved the first sender-deniable encryption with polynomial-length ciphertexts via indistinguishability obfuscation, transforming randomized algorithms into deniable versions through obfuscated programs generating fake randomness. Their scheme converts to receiver-deniable encryption via the generic transformation from [9], yielding a three-message protocol handling adversaries coercing either party separately, but not both simultaneously.

Impossibility results shaped understanding of deniable encryption’s limitations. Bendlin, Nielsen, Nordholt, and Orlandi [5] proved two-message receiver-deniable encryption impossible via information-theoretic compressibility arguments, establishing three-round necessity for bideniable schemes. Nielsen showed non-committing encryption permits only bounded public key reuse, implying deniable communication must be interactive. Dachman-Soled [21] established black-box construction limitations for sender-deniable encryption from simulatable public-key encryption.

¹ www.belenios.org

O’Neill, Peikert, and Waters [32] introduced multi-distributional bideniability under restricted settings, though their construction does not achieve standard bideniability where adversaries know the scheme and see all random coins. Unrestricted bideniable encryption remained open until Canetti, Park, and Poburinnaya [10] work.

Canetti, Park, and Poburinnaya [10] resolved the bideniable encryption existence question, where they presented a three-message protocol achieving bideniability and off-the-record deniability (protecting parties with inconsistent claimed states). Their construction requires subexponentially secure indistinguishability obfuscation and one-way functions, operates in the CRS model, addresses post-execution coercion, and explicitly identifies adaptive coercion and weaker assumption constructions as open problems.

Extensions to richer functionalities include: Apon, Fan, and Liu [4] constructed deniable attribute-based encryption from LWE for branching programs; De Caro, Iovino, and O’Neill [11] introduced deniable functional encryption denying both plaintexts and computed functions; Goldwasser, Klein, and Wichs [26] studied deniable editing of encrypted data with plausible deniability-preserving ciphertext modifications.

Our work builds most directly on the framework established by [10], but addresses the two main limitations they identified: the restriction to post-execution coercion and the reliance on subexponential security assumptions. By developing round-specific faking algorithms and a refined level system that operates under polynomial security, we achieve adaptive security where parties can be coerced at any point during protocol execution. Additionally, we strengthen the off-the-record deniability notion to require consistency between coordinated lies, and we demonstrate the first explicit application of fully deniable encryption to coercion-resistant voting with formal security analysis.

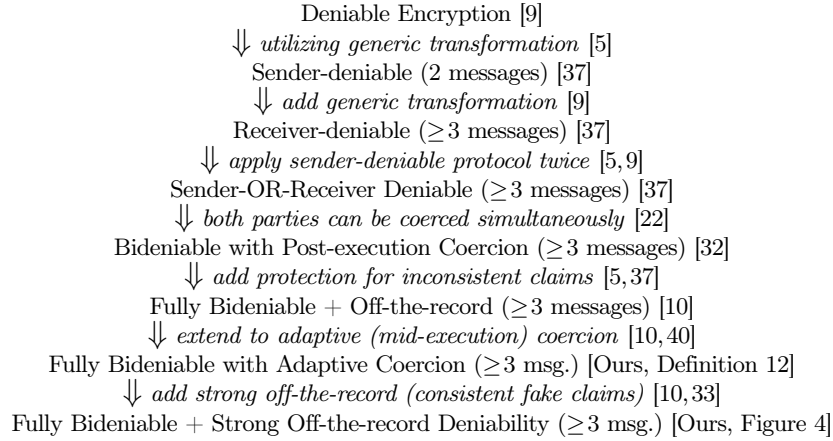


Fig. 1: Deniability hierarchy by compromised parties’ number, where arrows indicate the motivation inspired each contribution.

2 Preliminaries

Here, we establish the foundations required for our construction, beginning with hardness assumptions upon which our protocol relies, followed by a description of the primitives used as building blocks.

2.1 Computational Assumptions

Our construction of adaptively secure fully deniable encryption relies on three fundamental computational hardness assumptions with the following formalizations.

Learning with Errors (LWE) [36] For parameters $n, m, q \in \mathbb{N}$ and error distribution χ over $\mathbb{Z}_q$, no polynomial-time algorithm can distinguish $(\mathbf{A}, \mathbf{A}^T \mathbf{s} + \mathbf{e})$ from $(\mathbf{A}, \mathbf{u})$ where $\mathbf{A} \leftarrow \mathbb{Z}_q^{n \times m}$, $\mathbf{s} \leftarrow \mathbb{Z}_q^n$, $\mathbf{e} \leftarrow \chi^m$, $\mathbf{u} \leftarrow \mathbb{Z}_q^m$ with non-negligible advantage.

One-way Function (OWF) Hardness [27] A function $f: \{0,1\}^* \rightarrow \{0,1\}^*$ is one-way if it is polynomial-time computable but no polynomial algorithm can find any x' such that $f(x') = f(x)$ for uniformly random $x \leftarrow \{0,1\}^\lambda$ with non-negligible probability.

2.2 Building Blocks

The following describes the primitives that form the building blocks of our construction, and are organized according to their underlying hardness assumptions.

Definition 1 (Puncturable Pseudorandom Function [8]). Let $\lambda \in \mathbb{N}$ be a security parameter, and let $\mathcal{X}$ and $\mathcal{Y}$ be the input and output spaces. A puncturable pseudorandom function $\text{PPRF} = (\text{Setup}, \text{Eval}, \text{Puncture})$ consists of:

- $K_{\text{PRF}} \leftarrow \text{Setup}(1^\lambda)$: Outputs a PRF key K_{PRF} .
- $y \leftarrow \text{Eval}(K_{\text{PRF}}, x)$: On input key K_{PRF} and $x \in \mathcal{X}$, deterministically outputs $y \in \mathcal{Y}$. We write $\text{PPRF}(K, x)$ as shorthand for $\text{PPRF.Eval}(K, x)$.
- $K_{\{x^*\}} \leftarrow \text{Puncture}(K_{\text{PRF}}, x^*)$: On input key K_{PRF} and point $x^* \in \mathcal{X}$, outputs punctured key $K_{\{x^*\}}$ allowing evaluation on all $x \in \mathcal{X} \setminus \{x^*\}$.

Definition 2 (Chameleon Hash Function [30]). Let $p, q \in \mathbb{N}$ be primes with $p = kq + 1$ where q is sufficiently large, and let g be an element of order q in $\mathbb{Z}_p^*$. A chameleon hash function $\text{CH} = (\text{Gen}, \text{Hash}, \text{Coll})$ consists of:

- $(HK_R, CK_R) \leftarrow \text{Gen}(1^\lambda)$: Samples $x \leftarrow \mathbb{Z}_q^*$, computes $y = g^x \bmod p$, and outputs public hash key $HK_R = y$ and trapdoor $CK_R = x$.
- $h \leftarrow \text{Hash}(HK_R, m; r)$: On input hash key HK_R , message $m \in \mathbb{Z}_q^*$, and randomness $r \in \mathbb{Z}_q^*$, outputs $h = g^m y^r \bmod p$.
- $r' \leftarrow \text{Coll}(CK_R, HK_R, m', h, r)$: On input trapdoor $CK_R = x$, hash key HK_R , new message m' , hash value h , and randomness r where $h = \text{Hash}(HK_R, m; r)$, computes $r' = (m - m') \cdot x^{-1} - r \bmod q$ s.t. $\text{Hash}(HK_R, m; r) = \text{Hash}(HK_R, m'; r')$.

Definition 3 (ElGamal Encryption [25]). Let $\mathbb{G}$ be a cyclic group of prime order p with generator g . The ElGamal encryption scheme $\text{ElG} = (\text{Gen}, \text{Enc}, \text{Dec}, \text{Add})$ consists of:

- $(pk, sk) \leftarrow \text{Gen}(1^\lambda)$: Samples $x \leftarrow \mathbb{Z}_p$ uniformly, computes $h = g^x$, and outputs $pk = h$ and $sk = x$.
- $(c_1, c_2) \leftarrow \text{Enc}(pk, m; r)$: On input pk , message $m \in \mathbb{G}$, and randomness $r \leftarrow \mathbb{Z}_p$, computes $c_1 = g^r$ and $c_2 = m \cdot pk^r$, and outputs (c_1, c_2) .
- $m \leftarrow \text{Dec}(sk, (c_1, c_2))$: Outputs $m = c_2 \cdot c_1^{-sk}$.
- $(c_1, c_2) \leftarrow \text{Add}(\{(c_1^{(i)}, c_2^{(i)})\}_{i=1}^n)$: Outputs $(c_1, c_2) = (\prod_{i=1}^n c_1^{(i)}, \prod_{i=1}^n c_2^{(i)})$, which encrypts $\prod_{i=1}^n m_i$.

Definition 4 (Schnorr Signature [39]). Let $\mathbb{G}$ be a cyclic group of prime order p with generator g , and let $\mathcal{H}: \{0,1\}^* \rightarrow \mathbb{Z}_p$ be a hash function. The Schnorr signature scheme $\text{Sig} = (\text{Gen}, \text{Sign}, \text{Verify})$ consists of:

- $(vk, sk) \leftarrow \text{Gen}(1^\lambda)$: Samples $x \leftarrow \mathbb{Z}_p$ uniformly, computes $y = g^x$, and outputs verification key $vk = y$ and signing key $sk = x$.
- $(R, z) \leftarrow \text{Sign}(sk, m)$: On input $sk = x$ and message $m \in \{0,1\}^*$, samples $r \leftarrow \mathbb{Z}_p$, computes $R = g^r$, $c = \mathcal{H}(R \| m)$, and $z = r + c \cdot sk \bmod p$, and outputs (R, z) .
- $\{0,1\} \leftarrow \text{Verify}(vk, m, (R, z))$: Computes $c = \mathcal{H}(R \| m)$ and outputs 1 if $g^z = R \cdot vk^c$, else 0.

Definition 5 (Dual-Mode Commitment [35]). Let $n, m, q \in \mathbb{N}$ and χ be a distribution over $\mathbb{Z}_q$. A dual-mode commitment scheme $\text{DMLWE} = (\text{Setup}, \text{Com}, \text{Open}, \text{Equiv})$ operating in mode $\mu \in \{\text{equivocal}, \text{binding}\}$ consists of:

- $(pp, td) \leftarrow \text{Setup}(1^\lambda, \mu)$: Generates public parameters pp and trapdoor td based on $\text{LWE}(n, m, q, \chi)$, where td enables equivocation in **equivocal** mode and is $\perp$ in **binding** mode.
- $c \leftarrow \text{Com}(pp, m; r)$: On input pp , message $m \in \mathbb{Z}_q^n$, and randomness r , outputs commitment c .
- $\{0,1\} \leftarrow \text{Open}(pp, c, m, r)$: Outputs 1 if $c = \text{Com}(pp, m; r)$, else 0.
- $r_1 \leftarrow \text{Equiv}(td, c, m_0, m_1)$: On input trapdoor td , commitment $c = \text{Com}(pp, m_0; r_0)$, and messages m_0, m_1 , outputs randomness r_1 such that $c = \text{Com}(pp, m_1; r_1)$.

Definition 6 (Dual-Mode Public-Key Encryption [35]). Let $n, m, q \in \mathbb{N}$ and χ be a distribution over $\mathbb{Z}_q$. A dual-mode public-key encryption scheme $\text{DMLWE-PKE} = (\text{Setup}, \text{Gen}, \text{Enc}, \text{Dec}, \text{Explain})$ operating in mode $\mu \in \{\text{messy}, \text{decryption}\}$ consists of:

- $(pp, td) \leftarrow \text{Setup}(1^\lambda, \mu)$: Generates public parameters pp and trapdoor td , where td enables equivocation in **messy** mode and is $\perp$ in **decryption** mode.
- $(pk, sk) \leftarrow \text{Gen}(pp; r)$: Generates key pair (pk, sk) based on $\text{LWE}(n, m, q, \chi)$ using randomness r .
- $ct \leftarrow \text{Enc}(pk, m; s)$: On input pk , message $m \in \mathbb{Z}_q^n$, and randomness s , outputs ciphertext ct .
- $m \leftarrow \text{Dec}(sk, ct)$: Outputs message m . In **messy** mode, encryptions are statistically lossy.
- $s_1 \leftarrow \text{Explain}(td, pk, ct, m_0, m_1, s_0)$: On input trapdoor td , pk , ciphertext $ct = \text{Enc}(pk, m_0; s_0)$, and messages m_0, m_1 , outputs randomness s_1 such that $ct = \text{Enc}(pk, m_1; s_1)$.

Definition 7 (Non-Interactive Zero-Knowledge [34]). Let $\mathcal{R} \subseteq \{0,1\}^* \times \{0,1\}^*$ be an NP relation with language $\mathcal{L} = \{\Psi : \exists w \text{ s.t. } (\Psi, w) \in \mathcal{R}\}$. A non-interactive zero-knowledge proof system $\text{NIZK} = (\text{Setup}, \text{Prove}, \text{Verify}, \text{Sim})$ operating in mode $\mu \in \{\text{ZK}, \text{sound}\}$ consists of:

- $(\text{crs}, \text{td}) \leftarrow \text{Setup}(1^\lambda, \mu)$: Generates common reference string crs and trapdoor td , where td enables simulation in ZK mode and is $\perp$ in **sound** mode.
- $\pi \leftarrow \text{Prove}(\text{crs}, \Psi, w)$: On input crs , statement $\Psi \in \mathcal{L}$, and witness w where $(\Psi, w) \in \mathcal{R}$, outputs proof π .
- $\{0,1\} \leftarrow \text{Verify}(\text{crs}, \Psi, \pi)$: Outputs 1 if proof π is valid for statement Ψ , else 0.
- $\pi \leftarrow \text{Sim}(\text{crs}, \text{td}, \Psi)$: On input crs , trapdoor td , and statement Ψ (possibly $\notin \mathcal{L}$), outputs simulated proof π indistinguishable from honest proofs.

Interactive Deniable Encryption [9] Let $\mathcal{M}$ be a message space, an interactive deniable encryption scheme $\Pi = (\text{Setup}, P_1, P_2, P_3, \text{Dec}, \text{SFake}, \text{RFake})$ consists of:

- $\text{CRS} \leftarrow \text{Setup}(1^\lambda; r_{\text{setup}})$: On input security parameter 1^λ and randomness r_{setup} , the setup algorithm outputs a common reference string CRS .
- $\mu_1 \leftarrow P_1(s, m)$: On input sender randomness s and message $m \in \mathcal{M}$, the first protocol round outputs message μ_1 .
- $\mu_2 \leftarrow P_2(r, \mu_1)$: On input receiver randomness r and first-round message μ_1 , the second protocol round outputs message μ_2 .
- $\mu_3 \leftarrow P_3(s, m, \mu_1, \mu_2)$: On input sender randomness s , message m , and previous messages (μ_1, μ_2) , the third protocol round outputs message μ_3 .
- $m \leftarrow \text{Dec}(r, \text{tr})$: On input receiver randomness r and transcript $\text{tr} = (\mu_1, \mu_2, \mu_3)$, the decryption algorithm outputs message $m \in \mathcal{M} \cup \{\perp\}$.
- $s' \leftarrow \text{SFake}(s, m_{\text{true}}, m_{\text{fake}}, \text{tr}; \rho_S)$: On input sender's true randomness s , true message m_{true} , fake message m_{fake} , transcript tr , and faking randomness ρ_S , the sender faking algorithm outputs alternative randomness s' explaining tr as encoding m_{fake} .
- $r' \leftarrow \text{RFake}(r, m_{\text{true}}, m_{\text{fake}}, \text{tr}; \rho_R)$: On input receiver's true randomness r , true message m_{true} , fake message m_{fake} , transcript tr , and faking randomness ρ_R , the receiver faking algorithm outputs alternative randomness r' explaining tr as encoding m_{fake} .

3 Bideniability and Off-the-record Deniability

This section defines the security notions for interactive deniable encryption in the three-message setting. Here, we present these concepts formally by first specifying the correctness requirement, which ensures that honestly executed protocols decrypt to the intended plaintext with overwhelming probability.

Definition 8 (Correctness for Interactive Deniable Encryption). An interactive deniable encryption scheme $\Pi = (\text{Setup}, P_1, P_2, P_3, \text{Dec}, \text{SFake}, \text{RFake})$ for message space $\mathcal{M}$ satisfies correctness if there exists a negligible function $\nu(\lambda)$ such

that for at least a $(1-\nu(\lambda))$ fraction of setup randomness $r_{\text{setup}} \in \{0,1\}^{|r_{\text{setup}}|}$ and for all messages $m \in \mathcal{M}$, the following probability is overwhelming:

$$\Pr \left[\begin{array}{l} \text{CRS} \leftarrow \text{Setup}(1^\lambda; r_{\text{setup}}); \quad s \leftarrow \{0,1\}^{|s|}; \quad r \leftarrow \{0,1\}^{|r|}; \\ \mu_1 \leftarrow P_1(s, m); \quad \mu_2 \leftarrow P_2(r, \mu_1); \quad \mu_3 \leftarrow P_3(s, m, \mu_1, \mu_2); \\ \text{tr} \leftarrow (\mu_1, \mu_2, \mu_3); \quad m' \leftarrow \text{Dec}(r, \text{tr}); \quad \text{return } (m' = m) \end{array} \right] \geq 1 - \nu(\lambda).$$

We then present the Bideniability (Definition 9) and off-the-record deniability (Definition 10) security definitions, which capture the indistinguishability of real and fake internal states under various coercion scenarios.

Definition 9 (Bideniability). An interactive deniable encryption scheme Π for message space $\mathcal{M}$ satisfies bideniability if for any efficient adversary $\mathcal{A}$, the advantage $\text{Adv}_{\mathcal{A}}^{\text{BiDen}}(\lambda) = \left| \Pr[\text{Expt}_{\mathcal{A},0}^{\text{BiDen}}(1^\lambda) = 1] - \Pr[\text{Expt}_{\mathcal{A},1}^{\text{BiDen}}(1^\lambda) = 1] \right|$ is negligible in λ , where experiments $\text{Expt}_{\mathcal{A},0}^{\text{BiDen}}(1^\lambda)$ and $\text{Expt}_{\mathcal{A},1}^{\text{BiDen}}(1^\lambda)$ are defined in Figure 2.

$\text{Expt}_{\mathcal{A},0}^{\text{BiDen}}(1^\lambda)$	$\text{Expt}_{\mathcal{A},1}^{\text{BiDen}}(1^\lambda)$
1: $b \leftarrow 0$	1: $b \leftarrow 1$
2: $(m_0, m_1) \leftarrow \mathcal{A}(1^\lambda) \quad // \text{ where } m_0, m_1 \in \mathcal{M}$	2: $(m_0, m_1) \leftarrow \mathcal{A}(1^\lambda) \quad // \text{ where } m_0, m_1 \in \mathcal{M}$
3: $r_{\text{setup}} \leftarrow \{0,1\}^{ r_{\text{setup}} }$	3: $r_{\text{setup}} \leftarrow \{0,1\}^{ r_{\text{setup}} }$
4: $\text{CRS} \leftarrow \text{Setup}(1^\lambda; r_{\text{setup}})$	4: $\text{CRS} \leftarrow \text{Setup}(1^\lambda; r_{\text{setup}})$
5: $s^* \leftarrow \{0,1\}^{ s }; \quad r^* \leftarrow \{0,1\}^{ r }$	5: $s^* \leftarrow \{0,1\}^{ s }; \quad r^* \leftarrow \{0,1\}^{ r }$
6: $\mu_1^* \leftarrow P_1(s^*, m_0); \quad \mu_2^* \leftarrow P_2(r^*, \mu_1^*)$	6: $\mu_1^* \leftarrow P_1(s^*, m_1); \quad \mu_2^* \leftarrow P_2(r^*, \mu_1^*)$
7: $\mu_3^* \leftarrow P_3(s^*, m_0, \mu_1^*, \mu_2^*)$	7: $\mu_3^* \leftarrow P_3(s^*, m_1, \mu_1^*, \mu_2^*)$
8: $\text{tr}^* \leftarrow (\mu_1^*, \mu_2^*, \mu_3^*)$	8: $\text{tr}^* \leftarrow (\mu_1^*, \mu_2^*, \mu_3^*)$
9: $b' \leftarrow \mathcal{A}(\text{CRS}, m_0, m_1, s^*, r^*, \text{tr}^*)$	9: $\rho_S \leftarrow \{0,1\}^{ \rho_S }; \quad \rho_R \leftarrow \{0,1\}^{ \rho_R }$
10: return b'	10: $s' \leftarrow \text{SFake}(s^*, m_1, m_0, \text{tr}^*; \rho_S)$
	11: $r' \leftarrow \text{RFake}(r^*, m_1, m_0, \text{tr}^*; \rho_R)$
	12: $b' \leftarrow \mathcal{A}(\text{CRS}, m_0, m_1, s', r', \text{tr}^*)$
	13: return b'

Fig. 2: Bideniability Security Experiments for Interactive Deniable Encryption

Definition 10 (Off-the-Record Deniability). An interactive deniable encryption scheme Π for message space $\mathcal{M}$ satisfies the property of off-the-record deniability if for any efficient adversary $\mathcal{A}$, the advantage of the value $\text{Adv}_{\mathcal{A}}^{\text{OTR}}(\lambda) = \max_{i \neq j \in \{0,1,2\}} \left| \Pr[\text{Expt}_{\mathcal{A},i}^{\text{OTR}}(1^\lambda) = 1] - \Pr[\text{Expt}_{\mathcal{A},j}^{\text{OTR}}(1^\lambda) = 1] \right|$ is negligible in the λ , where experiments $\text{Expt}_{\mathcal{A},b}^{\text{OTR}}(1^\lambda)$ for $b \in \{0,1,2\}$ are defined in Figure 3.

$\text{Expt}_{A,0}^{\text{OTR}}(1^\lambda, m_0, m_1, m_2)$	$\text{Expt}_{A,1}^{\text{OTR}}(1^\lambda, m_0, m_1, m_2)$	$\text{Expt}_{A,2}^{\text{OTR}}(1^\lambda, m_0, m_1, m_2)$
1: $b \leftarrow 0$	1: $b \leftarrow 1$	1: $b \leftarrow 2$
2: $r_{\text{setup}} \leftarrow \{0,1\}^{ r_{\text{setup}} }$	2: $r_{\text{setup}} \leftarrow \{0,1\}^{ r_{\text{setup}} }$	2: $r_{\text{setup}} \leftarrow \{0,1\}^{ r_{\text{setup}} }$
3: $\text{CRS} \leftarrow \text{Setup}(1^\lambda; r_{\text{setup}})$	3: $\text{CRS} \leftarrow \text{Setup}(1^\lambda; r_{\text{setup}})$	3: $\text{CRS} \leftarrow \text{Setup}(1^\lambda; r_{\text{setup}})$
4: $s^* \leftarrow \{0,1\}^{ s }; r^* \leftarrow \{0,1\}^{ r }$	4: $s^* \leftarrow \{0,1\}^{ s }; r^* \leftarrow \{0,1\}^{ r }$	4: $s^* \leftarrow \{0,1\}^{ s }; r^* \leftarrow \{0,1\}^{ r }$
5: $\mu_1^* \leftarrow P_1(s^*, m_0); \mu_2^* \leftarrow P_2(r^*, \mu_1^*)$	5: $\mu_1^* \leftarrow P_1(s^*, m_1); \mu_2^* \leftarrow P_2(r^*, \mu_1^*)$	5: $\mu_1^* \leftarrow P_1(s^*, m_2); \mu_2^* \leftarrow P_2(r^*, \mu_1^*)$
6: $\mu_3^* \leftarrow P_3(s^*, m_0, \mu_1^*, \mu_2^*)$	6: $\mu_3^* \leftarrow P_3(s^*, m_1, \mu_1^*, \mu_2^*)$	6: $\mu_3^* \leftarrow P_3(s^*, m_2, \mu_1^*, \mu_2^*)$
7: $\text{tr}^* \leftarrow (\mu_1^*, \mu_2^*, \mu_3^*)$	7: $\text{tr}^* \leftarrow (\mu_1^*, \mu_2^*, \mu_3^*)$	7: $\text{tr}^* \leftarrow (\mu_1^*, \mu_2^*, \mu_3^*)$
8: $\rho_R \leftarrow \{0,1\}^{ \rho_R }$	8: $\rho_S \leftarrow \{0,1\}^{ \rho_S }$	8: $\rho_S \leftarrow \{0,1\}^{ \rho_S }; \rho_R \leftarrow \{0,1\}^{ \rho_R }$
9: $r' \leftarrow \text{RFake}(r^*, m_0, m_1, \text{tr}^*; \rho_R)$	9: $s' \leftarrow \text{SFake}(s^*, m_1, m_0, \text{tr}^*; \rho_S)$	9: $s' \leftarrow \text{SFake}(s^*, m_2, m_0, \text{tr}^*; \rho_S)$
10: $b' \leftarrow \mathcal{A}(\text{CRS}, m_0, m_1, m_2, s', r', \text{tr}^*)$	10: $b' \leftarrow \mathcal{A}(\text{CRS}, m_0, m_1, m_2, s', r', \text{tr}^*)$	10: $r' \leftarrow \text{RFake}(r^*, m_2, m_1, \text{tr}^*; \rho_R)$
11: return b'	11: return b'	11: $b' \leftarrow \mathcal{A}(\text{CRS}, m_0, m_1, m_2, s', r', \text{tr}^*)$
		12: return b'

Fig. 3: Interactive Deniable Encryption's Off-the-Record Deniability Experiments

Definition 11 (Fully Deniable Encryption [10]). Let $\mathcal{M}$ be a message space. A fully deniable encryption scheme satisfying both bideniability (Definition 9) and off-the-record deniability (Definition 10) under post-execution coercion is constructed from subexponentially secure indistinguishability obfuscation and one-way functions. The scheme $\Pi = (\text{Setup}, P_1, P_2, P_3, \text{Dec}, \text{SFake}, \text{RFake})$ consists of:

- $\text{CRS} \leftarrow \text{Setup}(1^\lambda; r_{\text{setup}})$: Outputs common reference string CRS containing six obfuscated programs implementing the protocol logic and faking capabilities.
- $\mu_i \leftarrow P_i(\cdot)$ for $i \in \{1, 2, 3\}$: Three protocol rounds exchanging messages μ_1, μ_2, μ_3 to transmit message $m \in \mathcal{M}$.
- $m \leftarrow \text{Dec}(r, \text{tr})$: On input receiver randomness r and transcript $\text{tr} = (\mu_1, \mu_2, \mu_3)$, outputs plaintext $m \in \mathcal{M} \cup \{\perp\}$.
- $s' \leftarrow \text{SFake}(s, m_{\text{true}}, m_{\text{fake}}, \text{tr}; \rho_S)$: Post-execution only, outputs fake sender randomness s' explaining completed transcript tr as encoding m_{fake} .
- $r' \leftarrow \text{RFake}(r, m_{\text{true}}, m_{\text{fake}}, \text{tr}; \rho_R)$: Post-execution only, outputs fake receiver randomness r' explaining completed transcript tr as encoding m_{fake} .

The faking algorithms in FDE (Definition 11) operate exclusively after protocol completion, leaving adaptive mid-execution coercion as an open problem explicitly identified in [10] work.

4 Strong Off-the-record Deniability

The off-the-record deniability notion presented in Definition 10 protects parties when their claimed plaintexts differ, addressing scenarios where the sender claims to have sent message m_0 while the receiver claims to have received message $m_1 \neq m_0$. However, this definition leaves unaddressed an important case: when both parties coordinate to claim a consistent fake plaintext that differs from the true message—for instance, when both parties are coerced simultaneously and choose to provide the same fabricated explanation to avoid suspicion or detection of their deception. While standard bideniability

ity (Definition 9) covers this scenario, it assumes both parties execute the protocol honestly and only lie afterward. Off-the-record deniability, conversely, considers arbitrary deviations but only guarantees protection when the parties' claims are inconsistent.

This gap becomes bigger in practical settings where parties may have opportunities to coordinate their responses under coercion, or where they face incentives to present a unified false narrative. In such cases, the adversary observes both parties consistently claiming the same fake plaintext, which should appear indistinguishable from a scenario where they consistently claim different fake plaintexts or where their claims conflict entirely. Without this guarantee, the mere consistency of the parties' claims could itself leak information about the true communication.

Therefore, we introduce Strong Off-the-record Deniability (SORD), which strengthens Definition 10 by adding a fourth indistinguishability case where both parties deviate with consistent claims. SORD ensures that no adversary can distinguish whether the observed consistency in the parties' claims reflects truth, coordination, or independent fabrication. Combined with bideniability, SORD protects all possible coercion scenarios and response patterns.

Definition 12 (Strong Off-the-Record Deniability). *Let $\pi = (\text{Setup}, P_1, P_2, P_3, \text{Dec}, \text{SFake}, \text{RFake})$ be a three-message interactive encryption scheme with message space $\mathcal{M}$. We strengthen Definition 10 by requiring indistinguishability under adaptive coercion (where parties may be coerced at any round during execution) and adding a fourth case where both parties deviate with consistent claims. We say that π satisfies strong off-the-record deniability (SORD) if no PPT adversary $\mathcal{A}$ wins with more than negligible advantage in the following game, for any $m_0, m_1, m_2 \in \mathcal{M}$.*

1. The challenger generates $\text{CRS} \leftarrow \text{Setup}(r_{\text{Setup}})$ for random r_{Setup} , and chooses $b \leftarrow \{0, 1, 2, 3\}$.
2. The challenger initializes the protocol execution with random s^*, r^* and tracks the current round $\rho \in \{1, 2, 3\}$.
3. $\mathcal{A}$ adaptively issues coercion queries (party, ρ) where $\text{party} \in \{\text{sender}, \text{receiver}\}$ and $\rho \in \{1, 2, 3, 4\}$ indicates the round at which coercion occurs (with $\rho = 4$ denoting post-execution coercion). Let τ_S and τ_R denote the coercion times for the sender and receiver, respectively (adaptive coercion phase).
4. Based on b and the coercion pattern (τ_S, τ_R) , the challenger responds as follows:
 - Sender truthful, Receiver deviates ($b=0$)
 - Generate $\text{tr}^* \leftarrow \pi(s^*, r^*, m_0)$
 - Upon sender coercion at round τ_S : give real partial state $\text{st}_S^{\tau_S}(s^*, m_0)$
 - Upon receiver coercion at round τ_R : Provide fake partial state $\text{st}_R^{\tau_R} \leftarrow \text{RFake}_{\tau_R}(r^*, m_0, m_1, \text{tr}_{<\tau_R}; \rho_R)$ where $\text{tr}_{<\tau_R}$ denotes the partial transcript up to round τ_R
 - Sender deviates, Receiver truthful ($b=1$)
 - Generate $\text{tr}^* \leftarrow \pi(s^*, r^*, m_1)$
 - Upon sender coercion at round τ_S : Provide fake partial state $\text{st}_S^{\tau_S} \leftarrow \text{SFake}_{\tau_S}(s^*, m_1, m_0, \text{tr}_{<\tau_S}; \rho_S)$
 - Upon receiver coercion at round τ_R : give real partial state $\text{st}_R^{\tau_R}(r^*, m_1)$
 - Both parties deviate with inconsistent claims ($b=2$)

- Generate $\text{tr}^* \leftarrow \pi(s^*, r^*, m_2)$
 - Upon sender coercion at round τ_S : Provide fake partial state $\text{st}_S^{\tau_S} \leftarrow \text{SFake}_{\tau_S}(s^*, m_2, m_0, \text{tr}_{<\tau_S}; \rho_S)$
 - Upon receiver coercion at round τ_R : Provide fake partial state $\text{st}_R^{\tau_R} \leftarrow \text{RFake}_{\tau_R}(r^*, m_2, m_1, \text{tr}_{<\tau_R}; \rho_R)$
 - Both parties deviate with consistent fake claims ($b=3$)
 - Generate $\text{tr}^* \leftarrow \pi(s^*, r^*, m_2)$
 - Upon sender coercion at round τ_S : Provide fake partial state $\text{st}_S^{\tau_S} \leftarrow \text{SFake}_{\tau_S}(s^*, m_2, m_0, \text{tr}_{<\tau_S}; \rho_S)$
 - Upon receiver coercion at round τ_R : Provide fake partial state $\text{st}_R^{\tau_R} \leftarrow \text{RFake}_{\tau_R}(r^*, m_2, m_0, \text{tr}_{<\tau_R}; \rho_R)$
5. After coercion, the protocol continues (if not yet complete) using the provided states. If a party is coerced mid-execution, the challenger continues the protocol execution using the fake state for that party while maintaining consistency with already-sent messages.
 6. $\mathcal{A}$ outputs $b' \in \{0,1,2,3\}$ and wins if $b' = b$.

Here, st_P^ρ denotes the internal state of party $P \in \{\text{sender}, \text{receiver}\}$ at round ρ , which includes all random coins used up to that point, all messages sent and received, and any intermediate computation results. The faking algorithms SFake_ρ and RFake_ρ are round-specific variants that can produce fake states at any round $\rho \in \{1,2,3,4\}$ that are consistent with the partial transcript observed so far. The advantage of $\mathcal{A}$ is defined as $|\Pr[\mathcal{A} \text{ wins}] - 1/4|$.

Our Definition 12 is stronger than Definition 10 and improves the off-the-record deniability from several aspects:

- it incorporates adaptive coercion at any point during protocol execution, not just post-execution;
- Definition 12 adds a fourth case ($b=3$) where both parties lie but their claims are consistent, which must be indistinguishable from the case where they lie with inconsistent claims;
- Our definition requires round-specific faking algorithms that can produce convincing fake states even mid-execution;
- it ensures that protocol continuation after mid-execution coercion remains indistinguishable across all cases.

A protocol satisfying both bideniability (Definition 9) and SORD (Definition 12) achieves the stronger form of deniability, which we term strongly fully deniable encryption in the next part.

5 Strongly Fully Deniable Encryption

This section demonstrates our deniable encryption protocol that maintains the three-message structure while achieving adaptive security under polynomial assumptions

through a combination of dual-mode building blocks with round-specific faking algorithms.

We present the construction of Strongly Fully Deniable Encryption (SFDE) in Figure 4 and Figure 5 that achieves both strong off-the-record deniability and bideniability under adaptive coercion, while relying only on polynomially-secure assumptions. Our protocol maintains the three-message structure and leverages a combination of dual-mode primitives with adaptive properties.

SFDE operates in three rounds under a CRS containing parameters for dual-mode commitments DMLWE, simulation-sound NIZK, chameleon hash CH, dual-mode PKE DMLWE-PKE, and puncturable PRF as PPRF. Setup generates parameters in `equivocal` and `messy` modes, providing trapdoors enabling equivocation. The DMLWE scheme offers computational hiding in `equivocal` mode with trapdoor-based opening to arbitrary values; DMLWE-PKE in `messy` mode produces statistically lossy ciphertexts explainable as encrypting any plaintext via the explanation trapdoor; PPRF generates session tags binding messages while maintaining pseudorandomness after puncturing, essential for security reductions.

In honest execution, sender computes (h_m, c_S, π_1) where $h_m = \text{CH.Hash}(HK_R, m; s_1)$, $c_S = \text{DC.Com}(pp_{\text{DC}}, (m, s_1); s_2)$, and π_1 proves knowledge of (m, s_1, s_2) . The receiver generates (pk, sk) , computes tag $\tau = \text{PPRF}(K_{\text{PRF}}, h_m \| c_S)$, commits $c_R = \text{DC.Com}(pp_{\text{DC}}, (\tau, \mu_1); r_2)$, and proves correct computation via π_2 . The sender encrypts $ct = \text{DPKE.Enc}(pk, (m, \mu_1, \mu_2, \tau); s_3)$ with consistency proof π_3 . The receiver decrypts, verifies transcript consistency, and outputs m .

SORD is achieved via round-adaptive faking algorithms enabling convincing fake randomness generation under adaptive coercion. When coerced, the sender invokes SFake_ρ computing the values of $s'_1 = \text{CH.Coll}(CK_R, HK_R, m_{\text{fake}}, h_m, s_1)$, $s'_2 = \text{DC.Equiv}(td_{\text{DC}}, c_S, (m_{\text{true}}, s_1), (m_{\text{fake}}, s'_1))$, along with the (post-round 3) $s'_3 = \text{DPKE.Explain}(td_{\text{PKE}}, pk, ct, (m_{\text{true}}, \dots), (m_{\text{fake}}, \dots), s_3)$. The receiver's RFake_ρ exploits that r does not encode m and DMLWE-PKE provides inherent deniability in messy mode, requiring minimal modifications via commitment equivocation. This independence ensures the unilateral generation of fake states without coordination, thereby satisfying off-the-record deniability despite inconsistent claims.

Here, Theorems 1, 2, 3 address the correctness, bideniability, and strong off-the-recordness of SFDE. The formal proofs for theorems are provided in Appendix A.

In the following, using Theorem 1, we address that the SFDE protocol satisfies correctness to make sure that honest executions decrypt to the intended plaintext with overwhelming probability.

Theorem 1. *The protocol $\Pi = (\text{Setup}, P_1, P_2, P_3, \text{Dec}, \text{SFake}, \text{RFake})$ described in Figure 4 satisfies correctness (Definition 8) if the underlying primitives PPRF, CH, DMLWE, DMLWE-PKE, and NIZK are correct.*

Now, in the following, we state that the SFDE protocol achieves bideniability, guaranteeing indistinguishability between real and fake internal states when both parties are coerced simultaneously.

Theorem 2. *The protocol Π described in Figure 4 satisfies bideniability (Definition 9) under the computational assumptions that PPRF is secure (Theorem 6), CH*

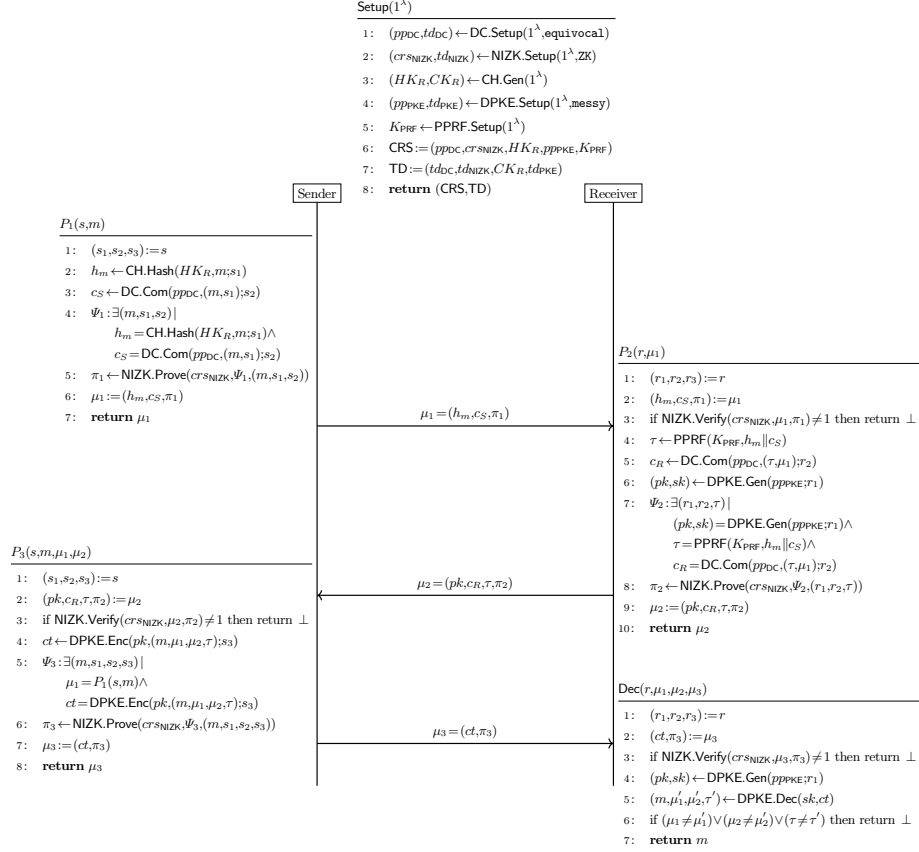


Fig. 4: Round-Adaptive Interactive Fully Deniable Encryption Protocol (SFDE)

satisfies collision resistance (Theorem 7), DMLWE satisfies equivocal hiding (Theorem 10), the DMLWE-PKE satisfies messy mode security (Theorem 11), and NIZK satisfies zero-knowledge (Theorem 12).

Finally, we address the fact that the SFDE protocol satisfies strong off-the-record deniability under adaptive coercion, which provides protection even when parties are coerced mid-execution and their claims are arbitrarily inconsistent or coordinated.

Theorem 3. *The protocol Π described in Figure 4 satisfies strong off-the-record deniability (Definition 12) under the same computational assumptions as Theorem 2, with the additional requirement that the faking algorithms SFake_ρ and RFake_ρ can be invoked at any round $\rho \in \{1, 2, 3, 4\}$ during protocol execution.*

SORD for SFDE extends the Bideniability argument by demonstrating that the lossy encryption property of dual-mode PKE in **messy** mode enables explaining

SFake₁ ($s, m_{\text{true}}, m_{\text{fake}}, \text{tr}_{<1}; \rho_S$)	SFake₂ ($s, m_{\text{true}}, m_{\text{fake}}, \text{tr}_{<2}; \rho_S$)
1: $(s_1, s_2, s_3) := s; (h_m, c_S, \pi_1) \leftarrow \text{tr}_{<1}$ 2: $s'_1 \leftarrow \text{CH.Coll}(CK_R, HK_R, m_{\text{fake}}, h_m, s_1)$ 3: $s'_2 \leftarrow \text{DC.Equiv}(td_{\text{DC}}, c_S, (m_{\text{true}}, s_1), (m_{\text{fake}}, s'_1))$ 4: $s'_3 := s_3; s' := (s'_1, s'_2, s'_3); \text{return } s'$	1: $(s_1, s_2, s_3) := s$ 2: $(\mu_1, \mu_2) := \text{tr}_{<2}$ 3: $(h_m, c_S, \pi_1) := \mu_1$ 4: $s'_1 \leftarrow \text{CH.Coll}(CK_R, HK_R, m_{\text{fake}}, h_m, s_1)$ 5: $s'_2 \leftarrow \text{DC.Equiv}(td_{\text{DC}}, c_S, (m_{\text{true}}, s_1), (m_{\text{fake}}, s'_1))$ 6: $s'_3 := s_3$ 7: $s' := (s'_1, s'_2, s'_3)$ 8: return s'
SFake₃ ($s, m_{\text{true}}, m_{\text{fake}}, \text{tr}_{<3}; \rho_S$)	RFake₂ ($r, m_{\text{true}}, m_{\text{fake}}, \text{tr}_{<2}; \rho_R$)
1: $(s_1, s_2, s_3) := s$ 2: $(\mu_1, \mu_2, \mu_3) := \text{tr}_{<3}$ 3: $(h_m, c_S, \pi_1) := \mu_1$ 4: $(pk, c_R, \tau, \pi_2) := \mu_2$ 5: $(ct, \pi_3) := \mu_3$ 6: $s'_1 \leftarrow \text{CH.Coll}(CK_R, HK_R, m_{\text{fake}}, h_m, s_1)$ 7: $s'_2 \leftarrow \text{DC.Equiv}(td_{\text{DC}}, c_S, (m_{\text{true}}, s_1), (m_{\text{fake}}, s'_1))$ 8: $s'_3 \leftarrow \text{DPKE.Explain}(td_{\text{PKE}}, pk, ct, (m_{\text{true}}, \mu_1, \mu_2, \tau), (m_{\text{fake}}, \mu_1, \mu_2, \tau), s_3)$ 9: $s' := (s'_1, s'_2, s'_3)$ 10: return s'	1: $(r_1, r_2, r_3) := r$ 2: $\mu_1 := \text{tr}_{<2}$ 3: $(h_m, c_S, \pi_1) := \mu_1$ 4: $r'_1 := r_1$ 5: $\tau \leftarrow \text{PPRF}(K_{\text{PRF}}, h_m c_S)$ 6: $c_R \leftarrow \text{DC.Com}(pp_{\text{DC}}, (\tau, \mu_1); r_2)$ 7: $r'_2 \leftarrow \text{DC.Equiv}(td_{\text{DC}}, c_R, (\tau, \mu_1), (\tau, \mu_1))$ 8: $r'_3 := r_3$ 9: $r' := (r'_1, r'_2, r'_3)$ 10: return r'
SFake₄ ($s, m_{\text{true}}, m_{\text{fake}}, \text{tr}; \rho_S$)	RFake₄ ($r, m_{\text{true}}, m_{\text{fake}}, \text{tr}; \rho_R$)
1: $s' \leftarrow \text{SFake}_3(s, m_{\text{true}}, m_{\text{fake}}, \text{tr}; \rho_S)$ 2: return s'	1: $r' := r$ 2: return r'
RFake₃ ($r, m_{\text{true}}, m_{\text{fake}}, \text{tr}_{<3}; \rho_R$)	
1: $r' := r$ 2: return r'	

Fig. 5: Round-Adaptive Faking Algorithms for Strongly Fully Deniable Encryption

any ciphertext as encrypting either consistent or inconsistent fake plaintexts indistinguishably. At the same time, round-specific faking algorithms maintain this indistinguishability under adaptive mid-execution coercion.

6 Voting using Strong Off-the-record Deniability

The SFDE protocol presented in the previous section, and here we demonstrate its applicability to practical systems by constructing a coercion-resistant electronic voting scheme. Coercion-resistant voting systems are essential to prevent vote-buying and intimidation attacks that undermine election integrity, as voters must be unable to prove how they voted even when coerced to reveal their credentials and protocol execution details, and this is where a tangible application of DE can be achieved. We build upon the Belenios voting system and prove that the resulting protocol achieves JCJ coercion resistance [29]. As a case study, we consider Belenios [19] e-voting system here, which can be seen as a variant of Helios [2] scheme. To our knowledge, no coercion-resistant variant of Belenios has been designed so far.

Definition 13 (Belenios [19]). *The Belenios voting scheme, based on labelled encryption LPKE $(\mathcal{E}', \Sigma'_{\mathcal{R}})$ where $\mathcal{E}' = (\text{KGen}', \text{Enc}', \text{Dec}', \text{Add})$ is homomorphic*

encryption, $\Sigma'_R = (P', V')$ proves vote validity, signature scheme $\Pi = (\text{SKGen}, \text{Sig}, \text{Vf})$, tally proof system $\Sigma_R = (P, V)$, hash function Hash , and revote policy Policy , consists of Belenios $[\text{Policy}] = (\text{Setup}, \text{Register}, \text{Vote}, \text{Valid}, \text{Tally}, \text{Verify}, \text{Publish}, \text{VerifyVote})$.

- $(pk, sk) \leftarrow \text{Setup}(1^\lambda)$: Generates election keys $(pk, sk) \leftarrow \text{KGen}'(1^\lambda)$.
- $(upk, usk) \leftarrow \text{Register}(id)$: For voter id , generates signing credential pair $(upk, usk) \leftarrow \text{SKGen}(1^\lambda)$ where upk is public and usk is secret.
- $b \leftarrow \text{Vote}(id, v, pk, usk)$: On input voter id , vote v , election key pk , and credential usk , derives upk from usk , computes $(c, \sigma) \leftarrow \text{Enc}'(pk, upk, v)$ where $\sigma \leftarrow P'((c, pk, upk), (v, r))$ proves vote validity, computes signature $s \leftarrow \text{Sig}(usk, (c, \sigma))$, and outputs ballot $b = (id, upk, (c, \sigma), s)$.
- $\{0,1\} \leftarrow \text{Valid}(\text{BB}, b, pk)$: On input ballot box BB , ballot $b = (id, upk, (c, \sigma), s)$, and public key pk , checks: (i) $\forall (x, y, \cdot, \cdot) \in \text{BB}: (x, y) = (id, upk) \vee (x \neq id \wedge y \neq upk)$ (no credential reuse across identities), (ii) $\forall' (c, \sigma, pk, upk) = 1$ (valid proof), (iii) $\text{Vf}(upk, (c, \sigma), s) = 1$ (valid signature), outputs 1 if all hold, else 0.
- $(r, \pi) \leftarrow \text{Tally}(\text{BB}, sk)$: On input stripped ballot box $\text{BB} = \{(upk, (c, \sigma), s)\}$ and secret key sk , filters valid ballots $\text{VBB} = \{(c, \sigma) \mid \forall' (c, \sigma, pk, upk) = 1 \wedge \text{Vf}(upk, (c, \sigma), s) = 1\}$, applies policy $C \leftarrow \text{Policy}(\text{VBB})$, homomorphically aggregates $\tilde{c} \leftarrow \text{Add}(\{c \mid (c, \sigma) \in C\})$, decrypts $r \leftarrow \text{Dec}'(sk, \tilde{c})$, computes $pbb \leftarrow \text{Publish}(\text{BB})$, generates proof $\pi \leftarrow P((pk, pbb, r), sk)$, and outputs (r, π) .
- $\{0,1\} \leftarrow \text{Verify}((pk, pbb, r), \pi)$: Returns $V((pk, pbb, r), \pi)$ verifying result r matches published board pbb .
- $pbb \leftarrow \text{Publish}(\text{BB})$: Returns $\{(b, \text{Hash}(b)) \mid b \in \text{Policy}(\text{BB})\}$.
- $\{0,1\} \leftarrow \text{VerifyVote}(b, pbb)$: Returns 1 iff $\text{Hash}(b) \in \{h \mid (\cdot, h) \in pbb\}$.

In the real-world web-application of classical Belenios, it uses ElGamal [25] (assuming DDH assumption holds in the group G ElGamal encryption scheme is not IND-CCA2 secure, but it is IND-CPA-secure [38]), and Schnorr signature [39] as core components.

6.1 Stronger Guarantees than Ballot Privacy in Voting

The Coercion Resistance property implies many other privacy guarantees, so it is a stronger notion than Receipt-freeness; i.e., Coercion Resistance [29] $\xRightarrow{[23]}$ Receipt-freeness [7] $\xRightarrow{[23]}$ Privacy with Malicious Board mb-BPRIV [20] $\xRightarrow{[20]}$ Ballot Privacy [6] $\xRightarrow{[16]}$ Individual Verifiability [18] $\xLeftrightarrow{[24]}$ Accountability [28]. Among all definitions of coercion resistance, Juels, Catalano & Jakobsson (JCJ) has been well-studied, and we consider the patched version of it [29] as Definition 14.

Definition 14 (Coercion Resistance [29]). Let $\Gamma = (\text{Setup}, \text{Register}, \text{Vote}, \text{Tally})$ be an election scheme, n_a, n_c and n_v be integers, and $\mathcal{D}$ be a distribution over $\{1, \dots, n_c, \phi, \lambda\}^{n_v}$. We say Γ satisfies JCJ with respect to $n_a, n_c, n_v, \mathcal{D}$, if for all probabilistic polynomial-time adversaries $\mathcal{A}$, there exists probabilistic polynomial-time algorithm $\mathcal{B}$, algorithm fakecred , and a negligible function negl , such that for all security parameters κ , we have:

$$|\Pr[\text{JCJ}(\Gamma, \mathcal{A}, n_a, n_c, n_v, \text{fakecred}, \mathcal{D}, \kappa) = 1] - \Pr[\text{JCJ}\text{-}\$(\Gamma, \mathcal{B}, n_a, n_c, n_v, \mathcal{D}, \kappa) = 1]| \leq \text{negl}(\kappa)$$

where games JCJ and JCJ-\$ are defined as follows in Figure 6, and algorithm fakecred takes a public key, a public credential, and a private credential as input, and outputs a (fake) private credential.

Game JSJ($\Gamma, \mathcal{A}, n_a, n_c, n_v, \text{fakecred}, \mathcal{D}, \kappa$)

<pre> 1: (pk, sk, mb, mc) $\leftarrow$ Setup(κ) 2: $V \leftarrow \mathcal{A}(pk, \kappa)$ 3: for $i \in [1, n_v]$ do 4: (pd_i, sd_i) $\leftarrow$ Register(pk, κ) 5: $L \leftarrow \{pd_1, \dots, pd_{n_v}\}$ 6: $M \leftarrow \{(i, sd_i) : i \in V \wedge i \in [1, n_v]\}$ 7: (j, v) $\leftarrow \mathcal{A}(M, L)$ 8: if $V \neq n_a \vee j \notin [1, n_v] \setminus V \vee v \notin [1, n_c] \cup \{\phi\}$ then 9: return 0 10: $bb_1 \leftarrow \emptyset$ 11: $\beta \leftarrow \mathcal{S}\{0, 1\}$ 12: if $\beta = 0$ then 13: if $v \neq \phi$ then 14: $b \leftarrow \text{Vote}(sd_j, pk, v, n_c, \kappa)$ 15: $bb_1 \leftarrow bb_1 \cup \{b\}$ 16: $sd' \leftarrow \text{fakecred}(pk, pd_j, sd_j)$ </pre>	<pre> 17: else 18: $sd' \leftarrow sd_j$ 19: for $i \in [1, n_v] \setminus (\{j\} \cup V)$ do 20: $v \leftarrow \mathcal{S}\mathcal{D}$ 21: if $v \neq \phi$ then 22: if $v = \lambda$ then 23: $v \leftarrow \mathcal{S}[1, n_c]$ 24: $sd_i \leftarrow \text{fakecred}(pk, pd_i, sd_i)$ 25: $b \leftarrow \text{Vote}(sd_i, pk, v, n_c, \kappa)$ 26: $bb_1 \leftarrow bb_1 \cup \{b\}$ 27: $bb_2 \leftarrow \mathcal{A}(sd', bb_1)$ 28: (v, pf) $\leftarrow \text{Tally}(sk, bb_1 \cup bb_2, L, n_c, \kappa)$ 29: $g \leftarrow \mathcal{A}(v, pf)$ 30: return ($\beta = g$) </pre>
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Fig. 6: The JSJ Game for Coercion Resistance

The JCJ-\$($\Gamma, \mathcal{B}, n_a, n_c, n_v, \mathcal{D}, \kappa$) game is identical to JCJ except:

- Line 13: $bb_2 \leftarrow \mathcal{B}(sd', M, |bb_1|)$ (receives bulletin board length)
- Line 14: $v \leftarrow \text{Ideal-Tally}(sk, bb_1 \cup bb_2, L, n_c, \kappa)$
- Line 15: $g \leftarrow \mathcal{B}(v)$ (no proof)

where Ideal-Tally tallies bb_1 normally and specially handles $bb_2 \setminus bb_1$ by counting ballots from corrupt voters without double-voting.

6.2 BeleniosCR: Coercion-resistant E-Voting

It has been proven that Belenios is ballot-private [17]. Also, it has been shown that Belenios can be receipt-free [12]. Here, we push this direction one step forward by showing that Belenios can be resistant to coercion as well in Figure 7, assuming apart from CH, let $\mathcal{H} : \{0, 1\}^* \rightarrow \{0, 1\}^\lambda$ be a collision-resistant hash function, and Policy(BB) denotes the latest valid ballot per voter in BB.

As Figure 7 shows, BeleniosCR extends the classical Belenios system with coercion resistance by integrating SFDE protocol into the ballot construction and verification mechanisms. The system preserves the fundamental structure of Belenios, including

homomorphic tallying and individual verifiability, while augmenting each ballot with deniable components derived from SFDE.

In BeleniosCR, setup generates public parameters $(pp_{\text{DC}}, crs_{\text{NIZK}}, HK_R, pk_{\text{enc}})$ for dual-mode commitments, NIZK, chameleon hash, and ElGamal encryption, enabling ballots satisfying both Belenios verifiability and deniability requirements for coercion resistance.

During voting, each voter constructs ballot $b = (\text{id}, \text{upk}, \mu_1, \tau, c_{\text{enc}}, \pi_2, s)$ by computing chameleon hash $h_v = \text{CH.Hash}(HK_R, v \parallel \text{upk}; r_1)$, dual-mode commitment $c_v = \text{DC.Com}(pp_{\text{DC}}, (v, r_1); r_2)$, NIZK proofs π_1, π_2 demonstrating well-formedness and vote validity, ElGamal ciphertext $c_{\text{enc}} = \text{ElG.Enc}(pk_{\text{enc}}, v; r_3)$ for homomorphic tallying, and Schnorr signature $s = \text{SS.Sign}(\text{usk}, (\mu_1, \tau, c_{\text{enc}}, \pi_2))$ binding components to the credential. This construction permits public verification and tally contribution while retaining capability to produce alternative randomness explaining the ballot as encoding a different vote.

Coercion resistance follows from FakeCred, enabling generation of fake randomness $\rho' = (r'_1, r'_2, r'_3)$ explaining ballot b as encoding coercer-demanded v_{fake} . When coerced, the voter invokes $\text{FakeCred}(pk, sk, \text{upk}, \rho, v_{\text{true}}, v_{\text{fake}})$, which computes $r'_1 = \text{CH.Coll}(CK_R, HK_R, v_{\text{fake}} \parallel \text{upk}, h_v, r_1)$ and $r'_2 = \text{DC.Equiv}(td_{\text{DC}}, c_v, (v_{\text{true}}, r_1), (v_{\text{fake}}, r'_1))$ while $r'_3 = r_3$ remains unchanged. Dual-mode primitives ensure ρ' is indistinguishable from randomness genuinely producing a ballot for v_{fake} , while c_{enc} continues encrypting v_{true} for the tally. This provides independent per-voter coercion protection, as coercers cannot distinguish fake from genuine randomness regardless of other voters' behavior.

We now establish the security of BeleniosCR in Theorem 4 by proving that it satisfies the JCJ coercion resistance (Definition 14) under standard assumptions.

Theorem 4. *Let n_a, n_c, n_v be integers representing the number of adversarially controlled voters, candidates, and total voters, respectively, and let $\mathcal{D}$ be a distribution over $\{1, \dots, n_c, \phi, \lambda\}^{n_v}$. The voting system BeleniosCR (Figure 7) satisfies JCJ coercion resistance (Definition 14) with respect to $n_a, n_c, n_v, \mathcal{D}$, under the assumptions that underlying SFDE protocol Π satisfies SORD (Theorem 3), ElG is IND-CPA-secure (Theorem 8), SS has EUF-CMA security (Theorem 9), and NIZK is sound (Theorem 12). Specifically, for any PPT adversary $\mathcal{A}$, there exists a PPT algorithm $\mathcal{B}$ and an algorithm $\text{BeleniosCR.FakeCred}$ such that:*

$$\begin{aligned} & |\Pr[\text{JCJ}(\text{BeleniosCR}, \mathcal{A}, n_a, n_c, n_v, \text{BeleniosCR.FakeCred}, \mathcal{D}, \kappa) = 1] \\ & - \Pr[\text{JCJ-}\$(\text{BeleniosCR}, \mathcal{B}, n_a, n_c, n_v, \mathcal{D}, \kappa) = 1]| \leq \text{negl}(\kappa). \end{aligned}$$

Proof. We prove coercion resistance by constructing a PPT simulator $\mathcal{B}$ that participates in the ideal experiment JCJ- $\$$ and produces a view computationally indistinguishable from the real experiment JCJ. The proof proceeds through several stages: simulator construction, indistinguishability argument, and tally consistency analysis.

We construct $\mathcal{B}$ that simulates the real JCJ game without knowing the actual votes of honest voters. $\mathcal{B}$ receives as input:

- Public parameters $pk = (pk_{\text{enc}}, pp_{\text{dc}}, crs_{\text{nizk}}, HK_R, K_{\text{prf}})$, $sk = (sk_{\text{enc}}, td_{\text{dc}}, td_{\text{nizk}}, CK_R)$ from BeleniosCR.Setup
- Credentials $M = \{(i, \text{usk}_i) : i \in V \wedge i \in [1, n_v]\}$ for adversarially controlled voters

BeleniosCR.Setup (1^λ) <pre> 1: $(pk_{enc}, sk_{enc}) \leftarrow \text{EIG.Gen}(1^\lambda)$ 2: $(pp_{dc}, td_{dc}) \leftarrow \text{DC.Setup}(1^\lambda, \text{equivocal})$ 3: $(crs_{nizk}, td_{nizk}) \leftarrow \text{NIZK.Setup}(1^\lambda, \mathbb{ZK})$ 4: $(HK_R, CK_R) \leftarrow \text{CH.Gen}(1^\lambda)$ 5: $K_{prf} \leftarrow \text{PPRF.Setup}(1^\lambda)$ 6: $pk \leftarrow (pk_{enc}, pp_{dc}, crs_{nizk}, HK_R, K_{prf})$ 7: $sk \leftarrow (sk_{enc}, td_{dc}, td_{nizk}, CK_R)$ 8: return (pk, sk)</pre>	BeleniosCR.Valid (BB, b, pk) <pre> 1: $(id, upk, \mu_1, \tau, c_{enc}, \pi_2, s) := b$ 2: $(h_v, c_v, \pi_1) := \mu_1$ 3: $(pk_{enc}, pp_{dc}, crs_{nizk}, HK_R, K_{prf}) := pk$ 4: $e_1 \leftarrow \text{NIZK.Verify}(crs_{nizk}, \psi_1, \pi_1)$ 5: $e_2 \leftarrow \text{NIZK.Verify}(crs_{nizk}, \psi_2, \pi_2)$ 6: $e_3 \leftarrow \text{SS.Verify}(upk, (\mu_1, \tau, c_{enc}, \pi_2), s)$ 7: $e_4 \leftarrow \forall (id', upk') \in BB. (id', upk') = (id, upk) \vee (id' \neq id \wedge upk' \neq upk)$ 8: return $(e_1 \wedge e_2 \wedge e_3 \wedge e_4)$</pre>
BeleniosCR.Register (id) <pre> 1: $(upk, usk) \leftarrow \text{SS.Gen}(1^\lambda)$ 2: return (upk, usk)</pre>	BeleniosCR.VerifyVote (b, pbb) <pre> 1: return $\mathcal{H}(b) \in \{h \mid (\cdot, h) \in pbb\}$</pre>
BeleniosCR.Vote (id, v, pk, upk, usk) <pre> 1: $(pk_{enc}, pp_{dc}, crs_{nizk}, HK_R, K_{prf}) := pk$ 2: $r_1, r_2, r_3 \leftarrow \\$(0, 1)^\lambda$ 3: $h_v \leftarrow \text{CH.Hash}(HK_R, v \ upk; r_1)$ 4: $c_v \leftarrow \text{DC.Com}(pp_{dc}, (v, r_1); r_2)$ 5: $\pi_1 \leftarrow \text{NIZK.Prove}(crs_{nizk}, \psi_1, (v, r_1, r_2))$ $\mid \psi_1 : \exists (v, r_1, r_2). h_v = \text{CH.Hash}(HK_R, v \ upk; r_1)$ $\wedge c_v = \text{DC.Com}(pp_{dc}, (v, r_1); r_2)$ 6: $\mu_1 \leftarrow (h_v, c_v, \pi_1)$ 7: $\tau \leftarrow \text{PPRF}(K_{prf}, id \ h_v \ c_v)$ 8: $c_{enc} \leftarrow \text{EIG.Enc}(pk_{enc}, v; r_3)$ 9: $\pi_2 \leftarrow \text{NIZK.Prove}(crs_{nizk}, \psi_2, (v, r_1, r_2, r_3))$ $\mid \psi_2 : \exists (v, r_1, r_2, r_3). \mu_1 = (h_v, c_v, \pi_1)$ $\wedge c_{enc} = \text{EIG.Enc}(pk_{enc}, v; r_3)$ $\wedge h_v = \text{CH.Hash}(HK_R, v \ upk; r_1)$ $\wedge c_v = \text{DC.Com}(pp_{dc}, (v, r_1); r_2)$ $\wedge v \in \mathcal{V}_{\text{valid}}$ 10: $s \leftarrow \text{SS.Sign}(usk, (\mu_1, \tau, c_{enc}, \pi_2))$ 11: $b \leftarrow (id, upk, \mu_1, \tau, c_{enc}, \pi_2, s)$ 12: $\rho \leftarrow (r_1, r_2, r_3)$ 13: return (b, ρ)</pre>	BeleniosCR.Tally (BB, sk) <pre> 1: $(sk_{enc}, td_{dc}, td_{nizk}, CK_R) := sk$ 2: $vbb \leftarrow []$ 3: for $b \in \text{Policy}(BB)$ do $(id, upk, \mu_1, \tau, c_{enc}, \pi_2, s) := b$ if BeleniosCR.Valid(BB, b, pk) then $vbb \leftarrow vbb \cup \{c_{enc}\}$ 4: $c_{total} \leftarrow \text{EIG.Add}(vbb)$ 5: $r \leftarrow \text{EIG.Dec}(sk_{enc}, c_{total})$ 6: $pbb \leftarrow \text{BeleniosCR.Publish}(BB)$ 7: $\pi_{\text{tally}} \leftarrow \text{NIZK.Prove}(crs_{nizk}, \psi_{\text{tally}}, sk_{enc})$ $\mid \psi_{\text{tally}} : \exists sk_{enc}, r = \text{EIG.Dec}(sk_{enc}, c_{total})$ $\wedge c_{total} = \text{EIG.Add}(\{c \mid (\cdot, \cdot, c, \cdot, \cdot) \in pbb\})$ 8: return (r, π_{tally})</pre>
BeleniosCR.Publish (BB) <pre> 1: $pbb \leftarrow \{(b, \mathcal{H}(b)) \mid b \in \text{Policy}(BB)\}$ 2: return pbb</pre>	BeleniosCR.FakeCred ($pk, sk, upk, \rho, v_{\text{true}}, v_{\text{fake}}$) <pre> 1: $(pk_{enc}, pp_{dc}, crs_{nizk}, HK_R, K_{prf}) := pk$ 2: $(sk_{enc}, td_{dc}, td_{nizk}, CK_R) := sk$ 3: $(r_1, r_2, r_3) := \rho$ 4: $h_v \leftarrow \text{CH.Hash}(HK_R, v_{\text{true}} \ upk; r_1)$ 5: $r'_1 \leftarrow \text{CH.Coll}(CK_R, HK_R, v_{\text{fake}} \ upk, h_v, r_1)$ 6: $c_v \leftarrow \text{DC.Com}(pp_{dc}, (v_{\text{true}}, r_1); r_2)$ 7: $r'_2 \leftarrow \text{DC.Equiv}(td_{dc}, c_v, (v_{\text{true}}, r_1), (v_{\text{fake}}, r'_1))$ 8: $c_{enc} \leftarrow \text{EIG.Enc}(pk_{enc}, v_{\text{true}}; r_3)$ 9: $r'_3 \leftarrow r_3$ 10: $\rho' \leftarrow (r'_1, r'_2, r'_3)$ 11: return ρ'</pre>
BeleniosCR.Verify ($(pk, pbb, r), \pi_{\text{tally}}$) <pre> 1: return $\text{NIZK.Verify}(crs_{nizk}, \psi_{\text{tally}}, \pi_{\text{tally}})$</pre>	

Fig. 7: BeleniosCR: Coercion-Resistant Belenios Voting Scheme Built upon Encryption Scheme EIG = (Gen, Enc, Dec, Add), Chameleon Hash CH = (Gen, Hash, Coll), Dual-mode Commitment DMLWE = (Setup, Com, Open, Equiv), Non-interactive Zero-knowledge NIZK = (Setup, Prove, Verify), Puncturable Pseudorandom Function PPRF = (Setup, Eval), and Schnorr Signature Sig = (Gen, Sign, Verify)

- Public credentials list $L = \{upk_1, \dots, upk_{n_v}\}$
- Challenge voter $j \in [1, n_v] \setminus V$, claimed vote $v \in [1, n_c] \cup \{\phi\}$
- Fake credential $usk' = sd'$ (real or fake depending on β)

$\mathcal{B}$ must simulate ballot generation for honest voters without knowing their true votes. For each honest voter $i \in [1, n_v] \setminus (\{j\} \cup V)$, $\mathcal{B}$ samples $v_i \leftarrow \mathcal{D}$ and proceeds based on v_i :

Case 1: $v_i = \phi$ (*abstention*). Generate no ballot for voter i .

Case 2: $v_i = \lambda$ (*coerced voter with fake credential*). This case simulates a coerced voter who uses a fake credential to cast a dummy ballot:

1. Sample true vote $v_i^{\text{true}} \leftarrow \mathcal{D}$ uniformly
2. Sample randomness $\rho_i = (r_1^{(i)}, r_2^{(i)}, r_3^{(i)}) \leftarrow \mathcal{D}^{3\lambda}$

3. Generate ballot components using the SFDE structure:

$$\begin{aligned}
h_{v_i} &\leftarrow \text{CH.Hash}(HK_R, v_i^{\text{true}} \parallel \text{upk}_i; r_1^{(i)}), c_{v_i} \leftarrow \text{DC.Com}(\text{pp}_{\text{dc}}, (v_i^{\text{true}}, r_1^{(i)}; r_2^{(i)}), \\
\pi_1^{(i)} &\leftarrow \text{NIZK.Prove}(\text{crs}_{\text{nizk}}, \Psi_1^{(i)}, (v_i^{\text{true}}, r_1^{(i)}, r_2^{(i)})), \mu_1^{(i)} := (h_{v_i}, c_{v_i}, \pi_1^{(i)}), \\
\tau_i &\leftarrow \text{PPRF}(K_{\text{prf}}, i \parallel h_{v_i} \parallel c_{v_i}), c_{\text{enc}}^{(i)} \leftarrow \text{ElG.Enc}(\text{pk}_{\text{enc}}, v_i^{\text{true}}; r_3^{(i)}), \\
\pi_2^{(i)} &\leftarrow \text{NIZK.Prove}(\text{crs}_{\text{nizk}}, \Psi_2^{(i)}, (v_i^{\text{true}}, r_1^{(i)}, r_2^{(i)}, r_3^{(i)})), \\
s_i &\leftarrow \text{SS.Sign}(\text{usk}_i, (\mu_1^{(i)}, \tau_i, c_{\text{enc}}^{(i)}, \pi_2^{(i)})), \\
b_i &:= (i, \text{upk}_i, \mu_1^{(i)}, \tau_i, c_{\text{enc}}^{(i)}, \pi_2^{(i)}, s_i).
\end{aligned}$$

where $v_i^{\text{fake}} \leftarrow \mathcal{S}[1, n_c]$ is sampled uniformly and represents the vote the coercer forces the voter to cast.

Case 3: $v_i \in [1, n_c]$ (honest vote). Generate ballot honestly by procedure $(b_i, \rho_i) \leftarrow \text{BeleniosCR.Vote}(i, v_i, \text{pk}, \text{upk}_i, \text{usk}_i)$. Collect all generated ballots in $bb_1 = \{b_i : i \in [1, n_v] \setminus (\{j\} \cup V), v_i \neq \phi\}$. For the challenge voter j , $\mathcal{B}$ receives $sd' = \text{usk}'$ which is either:

- Real credential usk_j (if $\beta = 1$ and $v = \phi$), or
- Fake credential generated by $\text{BeleniosCR.FakeCred}(\text{pk}, \text{sk}, \text{upk}_j, \rho_j, v_j^{\text{true}}, v)$ (if $\beta = 0$).

If $v \neq \phi$ and $\beta = 0$, a ballot for voter j was generated in previous stage and included in bb_1 . The credential sd' allows the adversary to "verify" that voter j cast a vote v (even though the true vote was v_j^{true}). $\mathcal{B}$ provides (sd', bb_1) to $\mathcal{A}$. In the real experiment, $\mathcal{A}$ receives (sd', M, bb_1) where M contains secret credentials for corrupt voters. In the ideal experiment, $\mathcal{B}$ receives $(sd', M, |bb_1|)$ (only the count, not the actual ballots). $\mathcal{A}$ generates additional ballots bb_2 using the corrupt voters' credentials in M . In the real game, $\mathcal{A}$ can see bb_1 and potentially tailor bb_2 accordingly. In the ideal game, $\mathcal{B}$ must generate bb_2 without seeing bb_1 . To maintain indistinguishability, $\mathcal{B}$ proceeds as follows:

1. $\mathcal{B}$ simulates $\mathcal{A}$ internally, providing $(sd', M, bb_1^{\text{sim}})$ where bb_1^{sim} consists of $|bb_1|$ ballots with dummy votes sampled uniformly from $[1, n_c]$
2. When $\mathcal{A}$ outputs bb_2 , $\mathcal{B}$ forwards bb_2 in the ideal experiment

The final bulletin board is $\text{BB} = bb_1 \cup bb_2$. The tally proceeds differently in real vs. ideal; the real experiment BeleniosCR.Tally computes:

$$\begin{aligned}
\text{vbb} &= \{c_{\text{enc}} : (\cdot, \cdot, \cdot, \cdot, c_{\text{enc}}, \cdot) \in \text{Policy}(\text{BB}), \text{BeleniosCR.Valid}(\text{BB}, b, \text{pk}) = 1\}, \\
c_{\text{total}} &\leftarrow \text{ElG.Add}(\text{vbb}), v \leftarrow \text{ElG.Dec}(\text{sk}_{\text{enc}}, c_{\text{total}}).
\end{aligned}$$

The ideal experiment Ideal-Tally computes:

- For ballots in bb_1 : Use actual encrypted votes (since these are honest voters' real votes)
- For ballots in $bb_2 \setminus bb_1$: Count only if from corrupt voters without double-voting

In both experiments, the tally correctly reflects the true votes of honest voters (not the fake claims) and the votes cast by corrupt voters. Now, we prove that $\mathcal{A}$'s views in real and ideal experiments are computationally indistinguishable via a sequence of hybrid games.

Hybrid $\mathcal{H}_0$: Real JCJ experiment. Challenge voter j casts vote v_j^{true} when $\beta=0, v \neq \phi$, and receives fake credential sd' making it appear as v .

Hybrid $\mathcal{H}_1$: Modify ballot generation for honest voters in Case 2 ($v_i = \lambda$). Instead of using true vote v_i^{true} , use dummy vote $v_i^{\text{dummy}} \leftarrow \$[1, n_c]$ uniformly. By SORD of the underlying SFDE protocol (Theorem 3), specifically the indistinguishability between cases $b=2$ and $b=3$, the fake credentials ρ'_i produce indistinguishable views regardless of whether they explain away v_i^{true} or v_i^{dummy} to v_i^{fake} : $|\Pr[\mathcal{H}_1] - \Pr[\mathcal{H}_0]| \leq n_v \cdot \text{Adv}_{\mathcal{A}'}^{\text{SORD}}(\lambda) \leq \text{negl}(\kappa)$, where $\mathcal{A}'$ is constructed from $\mathcal{A}$ to attack SORD.

Hybrid $\mathcal{H}_2$: Replace actual ballots bb_1 given to $\mathcal{A}$ with simulated ballots bb_1^{sim} containing dummy votes. By IND-CPA security of ElG (Theorem 8), ciphertexts $c_{\text{enc}}^{(i)}$ hide the encrypted votes: $|\Pr[\mathcal{H}_2] - \Pr[\mathcal{H}_1]| \leq |bb_1| \cdot \text{Adv}_{\mathcal{A}''}^{\text{IND-CPA}}(\lambda) \leq \text{negl}(\kappa)$. Additionally, by zero-knowledge of NIZK (Theorem 12), proofs $\pi_1^{(i)}, \pi_2^{(i)}$ can be simulated without witnesses: $|\Pr[\mathcal{H}_2'] - \Pr[\mathcal{H}_2]| \leq 2|bb_1| \cdot \text{Adv}_{\mathcal{A}'''}^{\text{ZK}}(\lambda) \leq \text{negl}(\kappa)$.

Hybrid $\mathcal{H}_3$: Ideal JCJ- $\$$ experiment. $\mathcal{B}$ generates bb_2 without seeing actual bb_1 . By construction, $\mathcal{H}_3 = \mathcal{H}_2'$. Combining all hybrids, we will have:

$$\begin{aligned} & |\Pr[\text{JCJ}(\text{BeleniosCR}, \mathcal{A}, \dots) = 1] - \Pr[\text{JCJ-}\$(\text{BeleniosCR}, \mathcal{B}, \dots) = 1]| = |\Pr[\mathcal{H}_0] - \Pr[\mathcal{H}_3]| \\ & \leq \sum_{i=0}^2 |\Pr[\mathcal{H}_{i+1}] - \Pr[\mathcal{H}_i]| \leq n_v \cdot \text{Adv}^{\text{SORD}}(\lambda) + |bb_1| \cdot (\text{Adv}^{\text{IND-CPA}}(\lambda) + 2\text{Adv}^{\text{ZK}}(\lambda)) \leq \text{negl}(\kappa). \end{aligned}$$

We verify that tallies match in real and ideal experiments. In both cases, we have: (I) Honest voters' ballots contain their true votes v_i^{true} (not v_i^{fake} claimed under coercion), (II) Corrupt voters' ballots contain votes chosen by $\mathcal{A}$, (III) The Policy function ensures only the latest valid ballots per voter are counted. The homomorphic property of ElG (Definition 18) ensures $\text{ElG.Dec}(\text{sk}_{\text{enc}}, \text{ElG.Add}(\text{vbb})) = \sum_{c \in \text{vbb}} v_c$, where v_c is the vote encrypted in ciphertext c . This matches the ideal tally function. We have now constructed a PPT simulator $\mathcal{B}$ and the sub-procedure $\text{BeleniosCR.FakeCred}$ such that the adversary $\mathcal{A}$'s advantage in distinguishing real from ideal experiments is negligible, combining all parts yields the above bound. Therefore, BeleniosCR satisfies JCJ coercion resistance (Definition 14). $\square$

7 Universal Verifiability vs. SFDE Trade-off

It has been a long-standing fact that in e-voting, it is impossible to achieve universal verifiability of the tally and unconditional privacy of the votes simultaneously [14]. Utilizing SFDE in voting schemes is not an exception, and although it brings strong privacy guarantees, the verifiability slightly diminishes, so we can not have SORD and universal verifiability [18] simultaneously.

Theorem 5. *Let the $\Gamma = (\text{Setup}, \text{Register}, \text{Vote}, \text{Tally}, \text{Verify}, \text{FakeCred})$ be a voting scheme with vote space $\mathcal{V}$. It is impossible for Γ to simultaneously satisfy:*

- (i) *Strong off-the-record deniability (Definition 12) with consistent fake claims, and*

(ii) *Universal verifiability, i.e., for all PPT adversaries $\mathcal{A}$ corrupting up to t voters:*

$$\Pr \left[\begin{array}{l} (\text{pk}, \text{sk}) \leftarrow \text{Setup}(1^\lambda); \text{BB} \leftarrow \mathcal{A}^{\text{Vote}(\cdot)}(\text{pk}); (r, \pi) \leftarrow \text{Tally}(\text{BB}, \text{sk}); \\ \text{Verify}((\text{pk}, \text{BB}, r), \pi) = 1 \wedge r \neq \text{TallyFunc}(\{v_i\}_{i \in H}) \end{array} \right] \leq \text{negl}(\lambda),$$

where H denotes the set of honest voters, v_i their true votes, and TallyFunc is the tally function.

Proof Sketch. We proceed by contradiction. Assume Γ satisfies both properties. Consider the following attack by letting voter V cast ballot b for candidate $v_{\text{real}} \in \mathcal{V}$ using randomness ρ . Let $v_{\text{fake}} \in \mathcal{V}$ with $v_{\text{fake}} \neq v_{\text{real}}$. After voting, V is coerced and uses FakeCred to produce fake randomness ρ' such that ballot b appears to encode v_{fake} . By SORD (Definition 12, case $b=3$), the tuple $(\text{pk}, b, v_{\text{fake}}, \rho')$ is computationally indistinguishable from a genuine vote for v_{fake} . The ballot box BB contains b along with other ballots. Universal verifiability requires that any verifier, given $(\text{pk}, \text{BB}, r, \pi)$, can check that r correctly reflects the votes in BB . To touch the contradiction, consider an external verifier $\mathcal{V}_{\text{ext}}$ who observes ballot b published on BB , tally result r with proof π , voter V 's claim: "I voted for v_{fake} using randomness ρ' ".

Since universal verifiability requires $\text{Verify}((\text{pk}, \text{BB}, r), \pi) = 1$ with overwhelming probability, and the tally must correctly count v_{real} (the actual vote), we have $r = \text{TallyFunc}(\dots, v_{\text{real}}, \dots)$. However, by SORD, the fake credentials $(\text{pk}, b, v_{\text{fake}}, \rho')$ are computationally indistinguishable from real credentials for voting v_{fake} . Therefore, $\mathcal{V}_{\text{ext}}$ should accept that b encodes v_{fake} , implying r should equal $\text{TallyFunc}(\dots, v_{\text{fake}}, \dots)$. This yields two mutually exclusive requirements:

$$\begin{aligned} r &= \text{TallyFunc}(\dots, v_{\text{real}}, \dots) && \text{(universal verifiability)} \\ r &\approx_c \text{TallyFunc}(\dots, v_{\text{fake}}, \dots) && \text{(SORD indistinguishability)} \end{aligned}$$

Since $v_{\text{real}} \neq v_{\text{fake}}$ and TallyFunc is efficiently computable, then $\text{TallyFunc}(\dots, v_{\text{real}}, \dots) \neq \text{TallyFunc}(\dots, v_{\text{fake}}, \dots)$ with overwhelming probability. This contradicts the computational indistinguishability required by SORD. Supposing both properties hold, construct a distinguisher $\mathcal{D}$ for SORD:

1. Receive (pk, b, v, ρ) from the SORD challenger (where $v \in \{v_{\text{real}}, v_{\text{fake}}\}$)
2. Construct $\text{BB} = \{b\}$ (single-voter case)
3. Compute $(r, \pi) \leftarrow \text{Tally}(\text{BB}, \text{sk})$
4. Run $\text{Verify}((\text{pk}, \text{BB}, r), \pi)$
5. If $r = \text{TallyFunc}(v_{\text{real}})$, output $v = v_{\text{real}}$; else output $v = v_{\text{fake}}$

By universal verifiability, $\mathcal{D}$ succeeds with probability $1 - \text{negl}(\lambda)$, violating SORD's indistinguishability requirement (Definition 12). Therefore, no voting scheme can satisfy both properties simultaneously. $\square$

8 Conclusion

We have introduced the first adaptively secure fully deniable encryption protocol based on polynomially secure assumptions, thus addressing the previously unresolved

issues of adaptive coercion and weaker primitives. The protocol we propose is able to recursively provide both bideniability and strong off-the-record deniability through round-specific fakes sub-procedures that guard parties at any point of execution from being forced to confess. By overcoming JCJ coercion resistance in electronic voting, our BeleniosCR construction showcases its real-world use. Yet, at the same time, our impossibility result brings forth the fact that the strong off-the-record deniability and the universal verifiability cannot stay together, thus pointing out another security properties trade-off. A future direction for this work includes finding alternative techniques to achieve coercion-resistance voting without SFDE. Another open problem is that it can be a constructive approach to design a homomorphic dual-mode LWE-based encryption scheme to remove the need for ElGamal in Belenios, which makes our protocol's applicability more efficient, since the dual-mode LWE-based scheme is limited to oblivious transfer, not homomorphic computation.

A Appendix

A.1 Properties of Building Blocks

Definition 15 (Pseudorandomness at Punctured Points). *A puncturable pseudorandom function PPRF (Definition 1) satisfies pseudorandomness at punctured points if for any probabilistic polynomial-time adversary $\mathcal{A}$ and all security parameters $\lambda \in \mathbb{N}$: $\text{Adv}_{\mathcal{A}}^{\text{PPRF}}(\lambda) := \left| \Pr[\text{Exp}_{\mathcal{A},0}^{\text{PPRF}}(1^\lambda) = 1] - \Pr[\text{Exp}_{\mathcal{A},1}^{\text{PPRF}}(1^\lambda) = 1] \right| \leq \text{negl}(\lambda)$, where experiment $\text{Exp}_{\mathcal{A},b}^{\text{PPRF}}(1^\lambda)$ is defined as:*

$$\begin{aligned} K_{\text{PRF}} &\leftarrow \text{PPRF.Setup}(1^\lambda); \quad x^* \leftarrow \mathcal{A}(1^\lambda); \quad K_{\{x^*\}} \leftarrow \text{PPRF.Puncture}(K_{\text{PRF}}, x^*); \\ y_0 &\leftarrow \text{PPRF.Eval}(K_{\text{PRF}}, x^*), \quad y_1 \leftarrow \mathcal{Y}; \quad b' \leftarrow \mathcal{A}(K_{\{x^*\}}, y_b); \quad \text{Output } b'. \end{aligned}$$

Theorem 6. *If the GGM pseudorandom generator $G : \{0,1\}^\lambda \rightarrow \{0,1\}^{2\lambda}$ is secure, then the GGM tree-based construction of PPRF satisfies pseudorandomness at punctured points (Definition 15). Specifically, for any PPT adversary $\mathcal{A}$ with advantage $\text{Adv}_{\mathcal{A}}^{\text{PPRF}}(\lambda)$, there exists a PPT algorithm $\mathcal{B}$ such that: $\text{Adv}_{\mathcal{A}}^{\text{PPRF}}(\lambda) \leq \log|\mathcal{X}| \cdot \text{Adv}_{\mathcal{B}}^G(\lambda) + \text{negl}(\lambda)$.*

Definition 16 (Correctness of Chameleon Hash). *A chameleon hash function $\text{CH} = (\text{Gen}, \text{Hash}, \text{Coll})$ (Definition 2) satisfies correctness if for all security parameters $\lambda \in \mathbb{N}$, all messages $m, m' \in \mathbb{Z}_q^*$, and all randomness $r \in \mathbb{Z}_q^*$:*

$$\Pr \left[\begin{array}{l} (HK_R, CK_R) \leftarrow \text{CH.Gen}(1^\lambda); \quad h \leftarrow \text{CH.Hash}(HK_R, m; r); \\ r' \leftarrow \text{CH.Coll}(CK_R, HK_R, m', h, r); \quad h' \leftarrow \text{CH.Hash}(HK_R, m'; r'); \quad h = h' \end{array} \right] = 1.$$

Definition 17 (Collision Resistance). *A chameleon hash function CH (Definition 2) satisfies collision resistance if for any probabilistic polynomial-time adversary $\mathcal{A}$ and all security parameters $\lambda \in \mathbb{N}$:*

$$\Pr \left[\begin{array}{l} (HK_R, CK_R) \leftarrow \text{CH.Gen}(1^\lambda); \quad (m, r, m', r') \leftarrow \mathcal{A}(HK_R); \\ \text{CH.Hash}(HK_R, m; r) = \text{CH.Hash}(HK_R, m'; r') \wedge (m, r) \neq (m', r') \end{array} \right] \leq \text{negl}(\lambda).$$

Theorem 7. *If the discrete logarithm problem in $\mathbb{Z}_p^*$ is hard, then the construction in Definition 2 satisfies collision resistance (Definition 17). Specifically, for any PPT adversary $\mathcal{A}$ finding collisions with probability $\epsilon(\lambda)$, there exists a PPT algorithm $\mathcal{B}$ solving the discrete logarithm problem such that $\text{Adv}_{\mathcal{B}}^{\text{DLog}}(\lambda) \geq \epsilon(\lambda) - \text{negl}(\lambda)$.*

Definition 18 (Homomorphic Property). *The scheme ElG (Definition 3) satisfies multiplicative homomorphism if for all messages $\{m_i\}_{i=1}^n \subseteq \mathbb{G}$ and corresponding ciphertexts $\{ct_i = (c_1^{(i)}, c_2^{(i)})\}_{i=1}^n : \text{ElG.Dec}(sk, \text{ElG.Add}(\{ct_i\}_{i=1}^n)) = \prod_{i=1}^n m_i$.*

Definition 19 (IND-CPA Security of ElGamal). *The ElGamal encryption scheme ElG (Definition 3) satisfies IND-CPA security if for any probabilistic polynomial-time adversary $\mathcal{A}$:*

$$\text{Adv}_{\text{ElG}, \mathcal{A}}^{\text{IND-CPA}}(\lambda) := \left| \Pr[\text{Exp}_{\mathcal{A}, 0}^{\text{IND-CPA}}(1^\lambda) = 1] - \Pr[\text{Exp}_{\mathcal{A}, 1}^{\text{IND-CPA}}(1^\lambda) = 1] \right| \leq \text{negl}(\lambda),$$

where: $(pk, sk) \leftarrow \text{ElG.Gen}(1^\lambda)$; $(m_0, m_1) \leftarrow \mathcal{A}(pk)$; $ct^* \leftarrow \text{ElG.Enc}(pk, m_b; r)$; $b' \leftarrow \mathcal{A}(ct^*)$; Output b' .

Theorem 8. *If the Decisional Diffie-Hellman (DDH) assumption holds in group $\mathbb{G}$, then ElGamal encryption (Definition 3) is IND-CPA secure (Definition 19). Specifically, for any PPT adversary $\mathcal{A}$ with advantage $\text{Adv}_{\text{ElG}, \mathcal{A}}^{\text{IND-CPA}}(\lambda)$, there exists a PPT algorithm $\mathcal{B}$ such that $\text{Adv}_{\text{ElG}, \mathcal{A}}^{\text{IND-CPA}}(\lambda) \leq \text{Adv}_{\mathcal{B}}^{\text{DDH}}(\lambda) + \text{negl}(\lambda)$.*

Definition 20 (EUF-CMA Security). *The Schnorr signature scheme Sig (Definition 4) satisfies existential unforgeability under chosen-message attack (EUF-CMA) if for any probabilistic polynomial-time adversary $\mathcal{A}$ making at most $Q = \text{poly}(\lambda)$ signing queries:*

$$\Pr \left[\begin{array}{l} (vk, sk) \leftarrow \text{Sig.Gen}(1^\lambda); (m^*, \sigma^*) \leftarrow \mathcal{A}^{\text{Sig}(sk, \cdot)}(vk); \\ \text{Sig.Verify}(vk, m^*, \sigma^*) = 1 \wedge m^* \notin \mathcal{Q} \end{array} \right] \leq \text{negl}(\lambda),$$

where $\mathcal{Q}$ is the set of messages queried to the signing oracle.

Theorem 9. *If the discrete logarithm problem is hard in group $\mathbb{G}$ and $\mathcal{H}$ is modeled as a random oracle, then the Schnorr signature scheme (Definition 4) is EUF-CMA secure (Definition 20). Specifically, for any PPT adversary $\mathcal{A}$ making at most Q_S signing queries and Q_H hash queries with success probability $\epsilon(\lambda)$, there exists a PPT algorithm $\mathcal{B}$ such that $\text{Adv}_{\mathcal{B}}^{\text{DLog}}(\lambda) \geq \frac{\epsilon(\lambda)}{Q_H} - \text{negl}(\lambda)$.*

Definition 21 (Mode Indistinguishability). *A dual-mode commitment scheme DMLWE (Definition 5) satisfies mode indistinguishability if for any PPT adversary $\mathcal{A}$ we can have the value of $|\Pr[(pp, td) \leftarrow \text{DMLWE.Setup}(1^\lambda, \text{equivocal}) : \mathcal{A}(pp) = 1] - \Pr[(pp, td) \leftarrow \text{DMLWE.Setup}(1^\lambda, \text{binding}) : \mathcal{A}(pp) = 1]| \leq \text{negl}(\lambda)$.*

Definition 22 (Computational Binding). *A dual-mode commitment scheme DMLWE (Definition 5) satisfies computational binding in **binding** mode if for any PPT adversary $\mathcal{A}$:*

$$\Pr \left[\begin{array}{l} (pp, \perp) \leftarrow \text{DMLWE.Setup}(1^\lambda, \text{binding}); (c, m_0, r_0, m_1, r_1) \leftarrow \mathcal{A}(pp); \\ \text{DMLWE.Open}(pp, c, m_0, r_0) = 1 \wedge \text{DMLWE.Open}(pp, c, m_1, r_1) = 1 \wedge m_0 \neq m_1 \end{array} \right] \leq \text{negl}(\lambda).$$

Theorem 10. If the $\text{LWE}_{n,m,q,\chi}$ assumption holds, then DMLWE (Definition 5) satisfies mode indistinguishability (Definition 21), computational binding in **binding** mode (Definition 22), and statistical hiding in **equivocal** mode. Specifically, for any PPT adversary $\mathcal{A}$, there exists a PPT algorithm $\mathcal{B}$ such that $\text{Adv}_{\text{DMLWE},\mathcal{A}}^{\text{Mode}}(\lambda) + \text{Adv}_{\text{DMLWE},\mathcal{A}}^{\text{Bind}}(\lambda) \leq \text{Adv}_{n,m,q,\chi,\mathcal{B}}^{\text{LWE}}(\lambda) + \text{negl}(\lambda)$.

Definition 23 (Mode Indistinguishability for PKE). DMLWE-PKE (Definition 6) satisfies mode indistinguishability if for any PPT adversary $\mathcal{A}$:

$$\left| \Pr[(pp, td) \leftarrow \text{Setup}(1^\lambda, \text{messy}) : \mathcal{A}(pp) = 1] - \Pr[(pp, td) \leftarrow \text{Setup}(1^\lambda, \text{decryption}) : \mathcal{A}(pp) = 1] \right| \leq \text{negl}(\lambda).$$

Definition 24 (IND-CPA Security of DMLWE-PKE). DMLWE-PKE (Definition 6) satisfies IND-CPA security if for any PPT adversary $\mathcal{A}$: $\text{Adv}_{\text{DMLWE-PKE},\mathcal{A}}^{\text{IND-CPA}}(\lambda) := |\Pr[\text{Exp}_{\mathcal{A},0} = 1] - \Pr[\text{Exp}_{\mathcal{A},1} = 1]| \leq \text{negl}(\lambda)$, where experiment $\text{Exp}_{\mathcal{A},b}(1^\lambda)$ is defined as:

$$(pp, \perp) \leftarrow \text{Setup}(1^\lambda, \text{decryption}); \quad (pk, sk) \leftarrow \text{Gen}(pp; r); \quad (m_0, m_1) \leftarrow \mathcal{A}(pp, pk); \\ ct^* \leftarrow \text{Enc}(pk, m_b; s); \quad b' \leftarrow \mathcal{A}(ct^*); \quad \text{Output } b'.$$

Theorem 11. If $\text{LWE}_{n,m,q,\chi}$ holds, then DMLWE-PKE (Definition 6) satisfies mode indistinguishability (Definition 23) and IND-CPA security (Definition 24). For any PPT adversary $\mathcal{A}$, there exists PPT $\mathcal{B}$ such that $\text{Adv}_{\mathcal{A}}^{\text{Mode}}(\lambda) + \text{Adv}_{\mathcal{A}}^{\text{IND-CPA}}(\lambda) \leq \text{Adv}_{n,m,q,\chi,\mathcal{B}}^{\text{LWE}}(\lambda) + \text{negl}(\lambda)$.

Definition 25 (Statistical Soundness). NIZK (Definition 7) satisfies statistical soundness in **sound** mode if for all (unbounded) adversaries $\mathcal{A}$:

$$\Pr \left[\begin{array}{l} (crs, \perp) \leftarrow \text{NIZK.Setup}(1^\lambda, \text{sound}); \quad (\Psi, \pi) \leftarrow \mathcal{A}(crs); \\ \text{NIZK.Verify}(crs, \Psi, \pi) = 1 \wedge \Psi \notin \mathcal{L} \end{array} \right] \leq \text{negl}(\lambda).$$

Definition 26 (Statistical Zero-Knowledge). NIZK (Definition 7) satisfies statistical zero-knowledge in **ZK** mode if there exists a simulator Sim s.t. for all unbounded adversaries $\mathcal{A}$ and all $(\Psi, w) \in \mathcal{R}$: $\Delta(\text{View}_{\text{real}}^{\mathcal{A}}(\Psi, w), \text{View}_{\text{sim}}^{\mathcal{A}}(\Psi)) \leq \text{negl}(\lambda)$, where:

$$\text{View}_{\text{real}}^{\mathcal{A}}(\Psi, w) := (crs, \pi, \mathcal{A}(crs, \pi)) \text{ with } (crs, td) \leftarrow \text{Setup}(1^\lambda, \text{ZK}), \pi \leftarrow \text{Prove}(crs, \Psi, w); \\ \text{View}_{\text{sim}}^{\mathcal{A}}(\Psi) := (crs, \pi', \mathcal{A}(crs, \pi')) \text{ with } (crs, td) \leftarrow \text{Setup}(1^\lambda, \text{ZK}), \pi' \leftarrow \text{Sim}(crs, td, \Psi).$$

Definition 27 (Computational Zero-Knowledge). NIZK (Definition 7) satisfies computational zero-knowledge in **sound** mode if there exists a simulator Sim such that for all $\mathcal{A}$ and all $(\Psi, w) \in \mathcal{R}$: $|\Pr[\mathcal{A}(\text{View}_{\text{real}}(\Psi, w)) = 1] - \Pr[\mathcal{A}(\text{View}_{\text{sim}}(\Psi)) = 1]| \leq \text{negl}(\lambda)$, where views are defined as in Definition 26 but with $(crs, td) \leftarrow \text{Setup}(1^\lambda, \text{sound})$.

Theorem 12. If $\text{LWE}_{n,m,q,\chi}$ holds, then the lattice-based NIZK construction (Definition 7) satisfies:

- (i) Mode indistinguishability between **ZK** and **sound** modes;

- (ii) Statistical soundness in **sound** mode (Definition 25) with soundness error $\leq 2^{-\lambda}$;
- (iii) Statistical zero-knowledge in **ZK** mode (Definition 26);
- (iv) Computational zero-knowledge in **sound** mode (Definition 27).

Specifically, for any PPT adversary $\mathcal{A}$, there exists PPT $\mathcal{B}$ such that $\text{Adv}_{\mathcal{A}}^{\text{Mode}}(\lambda) + \text{Adv}_{\mathcal{A}}^{\text{Comp-ZK}}(\lambda) \leq \text{Adv}_{n,m,q,\chi,\mathcal{B}}^{\text{LWE}}(\lambda) + \text{negl}(\lambda)$.

A.2 Proof of Theorem 1

Proof. We prove correctness by showing that for any message $m \in \mathcal{M}$, the probability that $\text{Dec}(r, \text{tr}) = m$ where $\text{tr} = (\mu_1, \mu_2, \mu_3)$ is the transcript of an honest execution, is overwhelming. Let $(s_1, s_2, s_3) := s$ and $(r_1, r_2, r_3) := r$ be the sender and receiver randomness, respectively. An honest execution proceeds as follows.

Round 1: Sender computes:

$$h_m \leftarrow \text{CH.Hash}(HK_R, m; s_1), c_S \leftarrow \text{DC.Com}(pp_{\text{DC}}, (m, s_1); s_2), \\ \pi_1 \leftarrow \text{NIZK.Prove}(crs_{\text{NIZK}}, \Psi_1, (m, s_1, s_2)), \mu_1 := (h_m, c_S, \pi_1).$$

Round 2: Receiver computes:

$$\tau \leftarrow \text{PPRF}(K_{\text{PRF}}, h_m \| c_S), c_R \leftarrow \text{DC.Com}(pp_{\text{DC}}, (\tau, \mu_1); r_2),$$

$$(pk, sk) \leftarrow \text{DPKE.Gen}(pp_{\text{PKE}}; r_1), \pi_2 \leftarrow \text{NIZK.Prove}(crs_{\text{NIZK}}, \Psi_2, (r_1, r_2, \tau)), \mu_2 := (pk, c_R, \tau, \pi_2).$$

Round 3: Sender computes:

$$ct \leftarrow \text{DPKE.Enc}(pk, (m, \mu_1, \mu_2, \tau); s_3), \pi_3 \leftarrow \text{NIZK.Prove}(crs_{\text{NIZK}}, \Psi_3, (m, s_1, s_2, s_3)), \mu_3 := (ct, \pi_3).$$

Decryption: Receiver computes: $(pk, sk) \leftarrow \text{DPKE.Gen}(pp_{\text{PKE}}; r_1)$, and $(m', \mu'_1, \mu'_2, \tau') \leftarrow \text{DPKE.Dec}(sk, ct)$. Now we analyze the failure probability:

Event E_1 : *NIZK proofs verify.* By completeness of NIZK, all honestly generated proofs π_1, π_2, π_3 verify with probability 1.

Event E_2 : *PKE decryption succeeds.* By correctness of DMLWE-PKE in decryption mode $\Pr[\text{DPKE.Dec}(sk, ct) \neq (m, \mu_1, \mu_2, \tau)] \leq \text{negl}(\lambda)$.

Event E_3 : *Consistency checks pass.* By determinism of PPRF (Theorem 6), $\tau = \text{PPRF}(K_{\text{PRF}}, h_m \| c_S)$ is uniquely determined, so $\tau' = \tau$. Similarly, $\mu'_1 = \mu_1$ and $\mu'_2 = \mu_2$ follow from the protocol structure. Therefore $\Pr[\text{Dec}(r, \text{tr}) \neq m] \leq \Pr[\neg E_1] + \Pr[\neg E_2 | E_1] + \Pr[\neg E_3 | E_1 \wedge E_2] \leq 0 + \text{negl}(\lambda) + 0 = \text{negl}(\lambda)$. Hence, $\Pr[\text{Dec}(r, \text{tr}) = m] \geq 1 - \text{negl}(\lambda)$, establishing correctness. $\square$

A.3 Proof of Theorem 2

Proof. Let $\mathcal{A}$ be a PPT adversary attacking bideniability with advantage $\epsilon(\lambda)$. We proceed with security via a sequence of games.

Game 0: This is $\text{Expt}_{\mathcal{A},0}^{\text{BiDen}}(1^\lambda)$ where the transcript encodes m_0 and the adversary receives honest randomness (s^*, r^*) .

Game 1: Modify CRS generation to use dual-mode primitives in equivocal/messy modes as $(pp_{\text{DC}}, td_{\text{DC}}) \leftarrow \text{DC.Setup}(1^\lambda, \text{equivocal})$, and the values $(pp_{\text{PKE}}, td_{\text{PKE}}) \leftarrow \text{DPKE.Setup}(1^\lambda, \text{messy})$, and $(crs_{\text{NIZK}}, td_{\text{NIZK}}) \leftarrow \text{NIZK.Setup}(1^\lambda, \text{ZK})$. By mode indistinguishability (Theorems 10, 11, 12) we have $|\Pr[\text{Game}_1] - \Pr[\text{Game}_0]| \leq \text{negl}(\lambda)$.

Game 2: Generate the transcript for m_1 instead of m_0 , but use trapdoors to explain it as m_0 : Compute the values of $h_m \leftarrow \text{CH.Hash}(HK_R, m_1; s_1)$ and $c_S \leftarrow \text{DC.Com}(pp_{\text{DC}}, (m_1, s_1); s_2)$. Use **Sim** to generate π_1 (zero-knowledge). Generate μ_2 honestly for the receiver. Compute $ct \leftarrow \text{DPKE.Enc}(pk, (m_1, \mu_1, \mu_2, \tau); s_3)$. Use **Sim** to generate π_3 . After generating the transcript, use faking algorithms:

$$\begin{aligned} s'_1 &\leftarrow \text{CH.Coll}(CK_R, HK_R, m_0, h_m, s_1), \quad s'_2 \leftarrow \text{DC.Equiv}(td_{\text{DC}}, c_S, (m_1, s_1), (m_0, s'_1)), \\ s'_3 &\leftarrow \text{DPKE.Explain}(td_{\text{PKE}}, pk, ct, (m_1, \mu_1, \mu_2, \tau), (m_0, \mu_1, \mu_2, \tau), s_3). \end{aligned}$$

Now we analyze indistinguishability through sub-hybrids.

Sub-hybrid 2.1: Chameleon hash equivocation. By the trapdoor property of CH (Theorem 7), the distribution of (h_m, s'_1) is identical to an honest computation for m_0 : $|\Pr[\text{Game}_{2.1}] - \Pr[\text{Game}_1]| = 0$.

Sub-hybrid 2.2: Commitment equivocation. By statistical hiding of DMLWE in equivocal mode (Theorem 10), c_S is statistically independent of the committed message: $|\Pr[\text{Game}_{2.2}] - \Pr[\text{Game}_{2.1}]| \leq \text{negl}(\lambda)$.

Sub-hybrid 2.3: Encryption equivocation. By the lossy encryption property of DMLWE-PKE in messy mode (Theorem 11), ct is statistically independent of the encrypted message: $|\Pr[\text{Game}_{2.3}] - \Pr[\text{Game}_{2.2}]| \leq \text{negl}(\lambda)$.

Sub-hybrid 2.4: Zero-knowledge simulation. By statistical zero-knowledge of NIZK (Theorem 12), simulated proofs are statistically indistinguishable from honest proofs: $|\Pr[\text{Game}_{2.4}] - \Pr[\text{Game}_{2.3}]| \leq \text{negl}(\lambda)$. Therefore: $|\Pr[\text{Game}_2] - \Pr[\text{Game}_1]| \leq \text{negl}(\lambda)$.

Game 3: This is $\text{Expt}_{\mathcal{A},1}^{\text{BiDen}}(1^\lambda)$ where the transcript encodes m_1 and fake randomness (s', r') are provided. By construction, $\text{Game}_3 = \text{Game}_2$. Thus, by combining all games we will have:

$$\text{Adv}_{\mathcal{A}}^{\text{BiDen}}(\lambda) = |\Pr[\text{Game}_0] - \Pr[\text{Game}_3]| \leq \sum_{i=0}^2 |\Pr[\text{Game}_{i+1}] - \Pr[\text{Game}_i]| \leq \text{negl}(\lambda).$$

□

A.4 Proof of Theorem 3

Proof. Let $\mathcal{A}$ be a PPT adversary attacking SORD with advantage $\epsilon(\lambda)$. The adversary must distinguish between four cases: ($b=0$) sender truthful, receiver deviates; ($b=1$) sender deviates, receiver truthful; ($b=2$) both deviate with inconsistent claims; ($b=3$) both deviate with consistent fake claims. We prove indistinguishability between all pairs of cases via hybrid arguments.

Indistinguishability between $b=0$ and $b=1$ follows from bideniability (Theorem 2). In $b=0$, the sender provides real state while the receiver fakes; in $b=1$, the sender fakes while the receiver provides real state. By symmetry of the protocol and bideniability, these are computationally indistinguishable:

$$|\Pr[\mathcal{A}(\text{View}_{b=0}) = 0] - \Pr[\mathcal{A}(\text{View}_{b=1}) = 1]| \leq \text{negl}(\lambda).$$

For Indistinguishability between $b=1$ and $b=2$ (Inconsistent Claims), consider case $b=1$ where the transcript encodes m_1 , the sender fakes to m_0 , receiver reveals

the true m_1 . In case $b=2$, transcript encodes m_2 , sender fakes to m_0 , receiver fakes to m_1 . We construct hybrids:

Hybrid H_0 : Case $b=1$ with $m_{\text{true}}=m_1$, sender fakes to m_0 .

Hybrid H_1 : Change transcript to encode m_2 , but use DPKE.Explain to make it appear as m_1 for receiver:

$$r'_3 \leftarrow \text{DPKE.Explain}(td_{\text{PKE}}, pk, ct, (m_2, \mu_1, \mu_2, \tau), (m_1, \mu_1, \mu_2, \tau), r_3).$$

By messy mode indistinguishability (Theorem 11): $|\Pr[H_1] - \Pr[H_0]| \leq \text{negl}(\lambda)$.

Hybrid H_2 : Receiver now fakes to m_1 (rather than revealing true m_1). Since ct statistically hides m_2 in messy mode, this change is undetectable: $|\Pr[H_2] - \Pr[H_1]| \leq \text{negl}(\lambda)$. Thus H_2 corresponds to case $b=2$: $|\Pr[\mathcal{A}(\text{View}_{b=1}) = 1] - \Pr[\mathcal{A}(\text{View}_{b=2}) = 2]| \leq \text{negl}(\lambda)$. The indistinguishability between $b=2$ and $b=3$ (Consistent Claims) is the core property of SORD. In $b=2$, sender claims m_0 and receiver claims m_1 (inconsistent). In $b=3$, both claim m_0 (consistent). Both transcripts encode the true message m_2 .

Hybrid G_0 : Case $b=2$ with transcript encoding m_2 , sender fakes to m_0 , receiver fakes to m_1 .

Hybrid G_1 : Modify receiver's fake to target m_0 instead of m_1 :

$$r'_3 \leftarrow \text{DPKE.Explain}(td_{\text{PKE}}, pk, ct, (m_2, \mu_1, \mu_2, \tau), (m_0, \mu_1, \mu_2, \tau), r_3).$$

The main point here is that ct is generated in messy mode, so by the lossy encryption property (Theorem 11), ct statistically hides m_2 . Therefore, explaining ct as encrypting either $(m_1, \dots)$ or $(m_0, \dots)$ produces identically distributed randomness:

$$\Delta(\text{DPKE.Explain}(td_{\text{PKE}}, pk, ct, (m_2, \mu_1, \mu_2, \tau), (m_1, \mu_1, \mu_2, \tau), r_3), \text{DPKE.Explain}(td_{\text{PKE}}, pk, ct, (m_2, \mu_1, \mu_2, \tau), (m_0, \mu_1, \mu_2, \tau), r_3)) \leq \text{negl}(\lambda).$$

Thus $|\Pr[G_1] - \Pr[G_0]| \leq \text{negl}(\lambda)$. Game G_1 corresponds exactly to case $b=3$: $|\Pr[\mathcal{A}(\text{View}_{b=2}) = 2] - \Pr[\mathcal{A}(\text{View}_{b=3}) = 3]| \leq \text{negl}(\lambda)$.

For the adaptive coercion, the faking algorithms SFake_ρ and RFake_ρ for $\rho \in \{1, 2, 3, 4\}$ maintain indistinguishability at each round by using the appropriate trapdoors:

- At round $\rho=1,2$: Use CK_R and td_{DC} to equivocate h_m and c_S .
- At round $\rho=3,4$: Additionally use td_{PKE} to equivocate ct .

Since the dual-mode primitives support statistical equivocation and all intermediate states are derived from these equivocable commitments, coercion at any round produces indistinguishable views. Combining all parts:

$$\text{Adv}_{\mathcal{A}}^{\text{SORD}}(\lambda) = \max_{i \neq j} |\Pr[\mathcal{A}(\text{View}_{b=i}) = i] - \Pr[\mathcal{A}(\text{View}_{b=j}) = j]| \leq \text{negl}(\lambda).$$

Therefore, the SFDE protocol satisfies SORD. $\square$

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